RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

SCHEME

Semester – I

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-101/1	Core	English Language	10	4	40	14	-	-	50
2	FC-101/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BMB 111	Core	Fundamental of Microbiology	20	8	80	27	50	17	150
4	BCH 111	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 111	Optional	Diversity of Microbes & Cryptogams	20	8	80	27	50	17	150
	BSZ 111	Optional	Invertebrates & Chordate	20	8	80	27	50	17	150
			Total	80	32	320	109	150	51	550

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc
Semester – I

Course	Subject	Subject Code
B.Sc(Microbiology)	English Language	FC-101/1

Unit-I

1. Amalkanti : Nirendranath Chakrabarti
2. Sita : Toru Dutt
3. Tryst with Destiny : Jawaharlal Nehru
4. Delhi in 1857 : Mirza Ghalib
5. Preface to the Mahabharata : C. Rajagopalachari
6. Where the Mind is Without Fear : Rabindranath Tagore
7. A Song of Kabir : Translated by Tagore
8. Satyagraha : M.K. Gandhi
9. Toasted English : R. K. Narayan
10. The Portrait of a Lady : Khushwant Singh
11. Discovering Babasaheb : Ashok Mahadevan

Unit-II

Comprehension

Unit-III

Composition and Paragraph Writing (Based on expansion of an idea).

Unit-IV

Basic Language Skills : Vocabulary – Synonyms, Antonyms, Word Formation, Prefixes and Suffixes, Words likely to be confused and Misused, Words similar in Meaning or Form, Distinction between Similar Expressions, Speech Skills

Unit-V

Basic Language Skills: Grammar and usage – The Tense Forms, Propositions, Determiners and Countable/Uncountable Nouns, Verb, Articles, Adverbs.

Prescribed Books: English Language and Indian Culture, Published by M.P. Hindi Grant Academy.

Note :- Eight questions to be set from unit-1 and four to be attempted.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc

Semester – I

Course	Subject	Subject Code
B.Sc(Microbiology)	**Development of Entrepreneurship** m|ferk fodkl	**FC-101/2**

Unit - I

Entrepreneurship- Definition, Characteristics and importance, Types and functions of an entrepreneur, merits of a good entrepreneur motivational factors of entrepreneurship.

उद्यमिता–परिभाषा, विशेषताऐं एवं महत्व, एक उद्यमी के प्रकार एवं कार्य, एक अच्छे उद्यमी के गुण, उद्यमिता अभिप्रेरणा ?kVdA

Unit - II

(a) Motivation to achieve targets and establishment of ideas. Setting targets and facing challenges. Resolving problems and creativity. Sequenced planning and guiding capacity, Development of self confidence.

(b) Communication skills, Capacity to influence, leadership.

v- y{; izkfIr dh ifj.kke ,oa fopkjksa dh LFkkiukA y{; fu/kkZj.k ,oa pqukSrh dk lkeukA
leL;k lek/kku ,oa l`tukRedrkA **क्रमबद्ध योजना एवं क्षमता की दिशाबद्धता।**
आत्मविश्वास का विकास।

c- lEizs"k.k dykA izHkkfor djus dh {kerkA usr`Ro

Unit - III

(a) Project Report - Evaluation of selected process. Detailed project report – Preparation of main part of project report pointing out necessary and viability.

(b) Selecting the form of Organization – Meaning and characteristics of sole Proprietorship, Partnership and cooperative committees, elements affecting selection of a form of an organisation.

(c) Economic management – Role of banks and financial institutions banking, financial plans, working capital-evaluation and management, keeping of accounts.

v- ifj;kstuk izfrosnu
pquh gbZ izfØ;k dk ewY;kadu
विस्तृत परियोजना प्रतिवेदन– आवश्यकता एवं प्रासंगिकता परियोजना प्रपत्र के प्रमुख भाग परियोजना प्रतिवेदन rS;kj djukA

c- laxBu ds izdkj dk p;u&,dkdh 0;olk;] lk>snkjh ,oa lgdkjh lfefr dk vFkZ ,**वं विशेषताएं संगठन के** p;u dks izHkkfor djus okys ?kVdA

l- vkfFkZd izca/ku

fofRr; lLFkku ,o cSad dh Hkwfedk] cSafdx] fofRr; ;kstuk] dk;Zdkjh iwath&eY;kadu rFkk izca/ku] ykxr o eY; fu/kkj.k rFkk ykHk dk eY;kadu] vkfFkZd ys[kk&tks[kk j[kukA

Unit - IV

(a) Production management. Methods of purchase. Management of movable assets/goods. Quality management. Employee management. Packaging.

(b) Marketing management. Sales and the art of selling. Understanding the market and market policy. Consumer management. Time management.

v- mRiknu dk izca/ku] [kjhnus ds rjhds] py lEifRr@eky dk izca/ku] xq.koRrk izca/ku] deZpkjh izca/ku] iSfdax

c- foi.ku izca/ku] fcØh ,oa cspus dh dyk] cktkj dh le> ,oa foi.ku uhfr] miHkksDrk izca/ku] le; izca/ku

Unit - V

(a) Role of regulatory institutions – district industry centre, pollution control board, food and drug administration, special study of electricity development and municipal corporation.

(b) Role of development organizations, khadi & village Commission/ Board, MP Finance Corporation, scheduled banks, MP Women's Economics Development Corporation.

(c) Self-employment-oriented schemes, Prime Minister's Employment schemes, Golden Jubilee Urban environment scheme, Rani Durgavati Self-Employment scheme, Pt. Deendayal Self-employment scheme.

(d) Various grant schemes – Cost-of-Capital grant, interest grant, exemption from entry tax, project report, reimbursement grant, etc.

(e) Special incentives for women entrepreneurs, prospects & possibilities.

(f) Schemes of M.P. Tribal Finance Development Corporation, schemes of M.P. Antyavasai Corporation, schemes of M.P. Backward Class and Minorities Finance Development Corporation.

1- fu;ked lLFkkvksa dh Hkwfedk&ftyk m|ksx dsUnz] iznw"k.k fuokj.k eaMy] [kk| ,oa vkS"kf/k प्रशासन, विद्युत विभाग तथा नगर निगम का विशेष अध्ययन।

2- विकासात्मक संस्थाओं की भूमिका, खादी एवं ग्रामीण आयोग/बोर्ड, मध्य प्रदेश वित्त निगम, अनुसूचित बैंक, मध्य प्रदेश का महिला आर्थिक विकास निगम।

3- Lojkstxkj eyd ;kstuk,sa & iz/kkuea=h jkstxkj ;kstuk] Lo.kZ t;arh 'kgjh jkstxkj ;kstuk] jkuh nqxkZorh Lojkstxkj ;kstuk] fnun;ky Lojkstxkj ;kstukA

4- विभिन्न अनुदान योजनाएँ– लागत पूँजी अनुदान, ब्याज अनुदान, प्रवेश कर से छूट, परियोजना प्रतिवेदन, प्रतिपूर्ति vuqnku vkfnA

5- महिला उद्यमियों हेतु विशेष प्रेरणk,a] lEHkkouk,sa ,oa leL;k,aA

6- मध्य प्रदेश आदिवासी वित्त विकास निगम की योजनाएँ , e-iz- vUR;kolk;h fuxe dh ;kstuk] e-iz- fiNMk oxZ ,oa vYila[;d foRr fodkl fuxe dh ;kstuk,aA

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Fundamental of Microbiology	BMB 111

Unit I

History, Taxonomy and Classification:
History and Scope of microbiology- Contributions of pioneers, Introduction to major groups of microorganisms and fields of Microbiology, Spontaneous generation *versus* biogenesis hypothesis, Whittaker's classification system of prokaryotes. Introduction to Bergey's manual of determinative and systematic classification, Bacterial nomenclature.

Unit II

Microscopy and Staining Techniques
Bright Field, Dark Field, Phase Contrast, Fluorescence and Scanning and Transmission Electron Microscopy, Stains and staining techniques- Stains and Dyes: classification and types, Types of staining- Simple (Monochrome, Negative), Differential (Gram and Acid fast).

Unit III

Morphology of Bacteria
Size, shape and arrangement of bacterial cells, Structures external to cell wall- Flagella, pili, capsule, sheath and prosthecae, Structures internal to cell wall- Cell membrane, nuclear material, cell wall (Protoplast and Spheroplast), spores, cytoplasmic inclusions, magnetosomes and plasmids.

Unit IV

Microbial Diversity
Bacteria with unusual properties- *Rickettsia, Chlamydia, Mycoplasma, Archaebacteria, Cyanobacteria, Actinomycetes,* Microbes of extreme environments– Adaptations and industrial importance of Thermophiles, Alkalophiles and Halophiles.

Unit V

Introduction to acellular forms of life
Introduction to viruses, viroids and prions, Structure of animal, plant and bacterial viruses, Classification and cultivation of viruses, Multiplication of bacterial viruses (lytic and lysogenic cycles).

Recommended Books (Semester-I)

1.Microbiology, Authors- Pelczar, Chan and Kreig.
2. Microbiology- an Introduction- (8th Edn), Authors- Tortora, G.J., Funke, B.R., Case, C.L.
3. General Microbiology, Authors- Stainer, Ingharam, Wheelis and Painter.
4. General Microbiology, Authors- Stainer RY. Ingharam JL. Wheelis ML. Painter PR
5. Biology of Microorganisms, Authors- Brock and Madigan.

6. Fundamental Principles of Bacteriology, Author- A.J. Salle.
7. Introduction to Microbiology, Authors- Ingraham and Ingraham.
8. Microbial Physiology, Authors- Moat and Foster.
9. Prokaryotic Development Authors- Brun, Y.V. and Shimkets, L.J. 2000, ASM Press

LIST OF EXPERIMENTS

1. Principles and working knowledge of instruments like autoclave, pH meter, incubator, hot air oven, centrifuge, microscope and colony counter.
2. Preparation of solid and liquid culture media and their sterilization.
3. Growth of bacteria on agar slant, agar stab, Petri plate and in broth.
4. Staining techniques- Simple staining, Gram staining, Negative staining, Endospore staining, Metachromatic granule staining, Spirochete staining.
5. Isolation of microorganisms by streak plate method.
6. Isolation of microorganisms by pour plate method.
7. Motility by hanging drop method.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus

Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	BASED ON INORGANIC, ORGANIC AND PHYSICAL CHEMISTRY	BCH-111

UNIT-I

Covalent Bond

Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridization and shapes of simple inorganic molecules and ions (BeF_2, BF_3, CH_4, PF_5, SF_6,IF_7 SO_4, ClO_4)Valence shell electron pair repulsion (VSEPR)theory to NH_3, H_3O^+, SF_4, CIF_3, ICl_2^- and H_2O. MO theory of hetero -nuclear (CO and NO) diatomic molecules, bond strength and bond energy.

UNIT-II

Gaseous States

Maxwell's distribution of velocities and energies (derivation excluded) Calculation of root mean square velocity, average velocity and most probable velocity. Collision diameter, co llision number, collision frequency and mean free path. Deviation of Real gases from ideal behavior. Derivation of Vander Waal's Equation of State.

UNIT-III

Critical Phenomenon:

Critical temperature, Critical pressure, Critical volume and their determination. PV isotherms of real gases,.

Liquid States -Structure of liquids. Properties of liquids – surface tension, viscosity vapour pressure and optical rotations and their determination.

Solid State

Classification of solids, Laws of crystallography – (i) Law of constancy of interfacial angles (ii) Law of rationality of indices(iii) Law of symmetry. Definition of unit cell & space lattice. X ray diffraction by crystals. Derivation of Bragg equation. Liquid crystals :,types of liquid crystals. Applications of liquid crystals.

UNIT-IV

Stereochemistry of Organic Compounds-I

Concept of isomerism. Types of isomerism. elements of symmetry, molecular chirality,enantiomers, stereogenic centre, , chiral and achiral molecules with two stereogeniccentres, diastereomers, mesocompounds, resolution of enantiomers, inversion, retention and racemization.

Stereochemistry of Organic Compounds-II

Sequence rules, R & S systems of nomenclature. Geometric isomerism determination of configuration of geometric isomers. E & Z system of nomenclature, isomerism conformational analysis of ethane and n-butane, conformations of cyclohexane, Newman projection and Sawhorse formulae.

UNIT- V

Mechanism of Organic Reactions

Curved arrow notation, drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking. Types of reagents – electrophiles and nucleophiles. Types of organic reactions. Energy considerations. Reactive intermediates carbocations, carbanions, free radicals, carbenes , arynes and nitrenes (formation, structure & stability).Assigning formal charges on intermediates and other ionic species.

Books:

1. Inorganic Chemistry, Huhee, Keiter and Keiter, Addition-Wesley, 1993.
2. Applied Mathematics for Physical Chemistery, J.R. Barante, Prenice Hall.
3. I. L. Levine, Quantum Chemistry , Prentice-Hall Inc., New Jersey.
4.. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
5. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
6. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International

Paper (Practical's)

Section-A (Inorganic)

Volumetric Analysis

1. Redox titrations: Determination of Fe^{2+} , $C_2O_4{}^{2-}$ (using $KMnO_4$,$K_2Cr_2O_7$)

2. Iodometic titrations: Determination of Cu^{2+} (using standard hyposolution).

3. Complexometric titrations: Determination of Mg^{2+} , Zn^{2+} by EDTA.

Section-B (Physical)

1. To determine the specific reaction rate of the hydrolysis of methyl acetate/ethyl acetatecatalyzed by hydrogen ions at room temperature.

2. To prepare arsenious sulphide sol and compare the precipitating power of mono-, bi – and trivalent anions.

Section – C (Organic)

1. Preparation and purification through crystallization ordistillation and ascertaining their purity through melting pointor boiling point
2. (i) Iodoform from ethanol (or acetone)
   (ii) *m*-Dinitrobenzne from nitrobenzene (use 1:2 conc. HNO_3 -H_2SO_4 mixture if fuming HNO_3 is not available)

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	**Diversity of Microbes & Cryptogams**	**BSB 111**

Unit-1

Viruses- Mycoplasma and Bacteria : characteristics of viruses and mycoplasma, general account of TMV and T4 bacteriophage. Bacterial structure, nutrition, reproduction and economic importance; general account of Cynobacteria.

Unit-2

Algae - General characters, classification and economic importance; important features and life history of Chlorophyceae- Volvox, Oedogonium, Charophyceae-Chara Xanthophyceae - Vaucheria, Phaeophyceae - Ectocarpus, Sargassum, Rhodophyceae - Polysiphonia.

Unit-3

Fungi- general characters, classification and economic importance, important features and life history of Mastigomycotina- Phytophthora, Zygomycotiana-Mucor. Asco mycotina -Aspergillus, Peziza, Basidomycotina- Puccinia, Deurteromycotina- Cercospora, Colletotrichum, general account of Lichens.

Unit-4

Bryophyta - Classification, study of morphology, anatomy, reproduction of Hepaticopsida, Riccia, Marchantia, Anthrocerotopsida Anthoceros, Bryopsida- Polytrichum

Unit-5

Pteridophyta - Important characters and classification. Stelar organization. Morphology and anatomy of Rhynia. Structure, anatomy and reproduction in Lycopodium, Selaginella, Equisetum and Marsilea.

Suggested Books :

1.G.M. Smith 1971 Cryptogamic Botany. Vol - I Algae & Fungi Tata McGrraw Hill Pub. Co. New Delhi.

2.G.M. Smith 1971 Cryptogamic Botany. Vol -II Bryophytes & Pteridophytes. Tata McGrraw Hill Pub. Co. New Delhi.

3.O.P.Sharma,1992. Text book of Thallophyta McGrraw Hill Pub. Co.

4.O.P.Sharma,1990. Text book of Pteridophyta McMillan India Ltd .

5.P.D.Sharma 1991. The Fungi. rastogi & Co. Meerut.

6.H.C. Dubey.1990. an introduction of Fungi.Vikas Pub. house pvt.ltd.

7.P.Puri 1980. Bryophyta Atma Ram & Sons, Delhi.

8.A.Clifton.1958. Introduction to the Bacteria. Mcgrew Hillpub. Co.New delhi.

Practical Work : Semester-I

Scheme of practical examination **Marks:**

Algae / Fungi 05 Brophyta 10 Pteridophyta 10

Plant disease 05 Spoting 10 Sessional 10

Total 50

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	INVERTEBBRATA & CHORDATA	BSZ 111

Unit-1

A brief introduction and nomenclature - Level of Organization. Phylum Protozoa: General characters- Classifications. Type study-Paramecium-Structure and Reproduction. General Topic: Protozoan Diseases.

Unit -2

Phylum Porifera: General characters- Type study- Ascon- Cellular structure. Phylum Coelenterata: Classification- Type study- Aurelia- Structure and life history. General Topic: Canal system in Sponges, Polymorphism in Coelenterates. Phylum Mollusca: General characters- Type study – Unio (Lamellidens) - External morphology and digestive system. General Topic: Mouth parts of Insects- Economic Importance of Mollusca.

Unit -3

Phylum Platyhelmenthes: General Characters- Classification- Type study- Liver fluke- Structure and Reproduction. Phylum Annelida: General characters- Type study- Nereis- External morphology and reproduction. General Topic: Helminth Parasites in Man.

Unit-4

Introduction- Type Study: Amphioxus- external characters, digestive, excretory, respiratory, and circulatory systems.

Class: Pisces, General characters – Type Study: Scoliodon-External characters, Digestive, Excretory, Respiratory and Circulatory Systems- Structure of Brain-Sense organs and Reproductive system. General Topic: Accessory respiratory organs in fishes.

Unit-5

Class : Amphibia :General characters and classification -Type Study : Frog –External characters, Digestive ,Respiratory, Circulatory and Reproductive systems - Structure of Brain.

Class: Reptelia: General characters- Type study –Calotes- External characters-Digestive, Respiratory, Circulatory and Reproductive systems- Structure of Brain. General Topic: Identification of Poisonous and Non- Poisonous snakes. Golden age of Reptiles.

References

1. Agarwal V.K (2000) Invertebrate Zoology- S.Chand Company.
2. Barnes R.D (1987) Invertebrate Zoology- Saunders College Publications.
3. Barrington E.J (1981) Invertebrate Structure and Function. ELBS Editions.
4. Ekambaranatha Iyer (1993) Manual of Zoology Volume I Invertebrata.

5. Kotpal R.L (2003) Modern text book of Zoology- Rostogi Publications, Meerut.
6. Ekambaranatha Iyer (1993) Manual of Zoology Vol.II, Viswanathan (printers& publishers) Chennai.
7. Jordon, E.L & Verma, P.S. (2000) Chordate Zoology, S.Chand & Co, New Delhi.
8. Newman H.H., Chordata, McMilan publishers.

SEMESTER-I
PRACTICLE-I
INVERTEBATA AND CHORDATA

I. Major Practicals:
Cockroach-Nervous, digestive, Reproductive system
Prawn-Nervous system

II. Minor Practicals:
Prawn –Appendeges
Mouth parts –Honey Bee, Mosquito, and Cockroach.

III. Spottors:
a) Classify giving reations:
Entamoeba, Paramecium, Leucosolenia, Hyalonema, Aurelia, Obelia, Taenia, Ascaris, Earthworm, Nereis, Penaeus, Freshwater mussel, Starfish, Cockroach, Amphioxus, Salpa, Frog, Cobra, Pigeon, Rabbit.
b) Draw Labelled Sketch:
T.S. of Taenia ,T.S .of Fasciola , Ephyra larva , Nauplius larva , Zoea larva ,Quill feather , Frog –Pectoral girdle ,Pigeon –Pelvic girdle .
c) Biological Significance:
Sponge –Gemmule, Physalia, Leech, Limulus, Bipinnaria, Ascidian tadpole larva, Ichthiophis, Peripatus.
d) Relate structure and function
Taenia –Scolex , Nereis – Parapodium, Peneus –Petasma, Star fish –Tube feet (ventral view), Echenies, Draco, Bat.
e) Comment on Respiratory /skeletal/ dentition of the following
Star fish, Synsacrum, Dentition of Rabbit and Dog.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

SCHEME

Semester – II

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-201/1	Core	हिन्दी भाषा संरचना	10	4	40	14	-	-	50
2	FC-201/2	Core	Environment	10	4	40	14	-	-	50
3	BMB 121	Core	MICROBIAL PHYSIOLOGY & MICROBIAL GENETICS	20	8	80	27	50	17	150
4	BCH 121	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 121	Optional	Cell Biology & Genetics	20	8	80	27	50	17	150
	BSZ 121	Optional	Cell Biology & Genetics	20	8	80	27	50	17	150
			Total	80	32	320	109	150	51	550

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc
Semester – II

Course	Subject	Subject Code
B.Sc(Microbiology)	हिन्दी भाषा संरचना	**FC-201/1**

इकाई –1

भारत वंदना (काव्य) : सूर्यकांत त्रिपाठी 'निराला'
जाग तुझकों दूर जाना : सुश्री महादेवी वर्मा
स्वतंत्रता पुकारती (काव्य) : जयशंकर 'प्रसाद'
हम अनिकेतन (काव्य) : बालकृष्ण शर्मा 'नवीन'
भाषा की महत्ता और उसके विविध रूप

भाषा–कौशल

इकाई –2

करूणा (निबंध) : आचार्य रामचन्द्र शुक्ल
समन्वय की प्रक्रिया (निबंध) : रामधारी सिंह 'दिनकर'
बिच्छी बुआ (कहानी) : डॉ. लक्ष्मण विष्ट 'बटरोही'
अनुवाद : परिभाषा प्रकार, महत्व, विशेषताएं
हिन्दी की शब्द–संपदा
परिभाषिक शब्दावली

इकाई–3

विलायत पहुँच ही गया (आत्मकथांश) : महात्मा गांधी
अफसर (व्यंग्य) : शरद जोशी
तीर्थयात्रा (कहानी) : डॉ. मिथिलेश कुमारी मिश्र
मकड़ी का जाला (व्यंग्य) : डॉ. रामप्रकाश सक्सेना
वाक्य–संरचनाः तत्सम, तद्भव देशज विदेशी

इकाई–4

अप्प दीपो भव (वक्तृत्व कला) : स्वामी श्रद्धानन्द
भारत का सामासिक व्यक्तित्व (प्रस्तावना) : जवाहरलाल नेहरू
पत्र मैसूर के महाराजा को (पत्र–लेखन) : स्वामी विवेकानन्द
बनी रहेगी किताबें (आलेख) : डॉ. सुनीता रानी घोष
पत्र–लेखन : महत्व और उसके विविध रूप
सड़क पर दौड़ते ईहा मृग (निबंध) डॉ. श्यामसुन्दर दुबे

इकाई–5

योग की शक्ति (डायरी) : डॉ. हरिवंशराय बच्चन
कोश के अखाड़े में कोई पहलवान नही उतरता (साक्षात्कार) : भाषाविद् डॉ. हरदेव बाहरी से प्रो. त्रिभुवननाथ शुक्ल
नीग्रो सैनिक से भेंट (यात्रा–संस्मरण) : डॉ. देवेन्द्र सत्यार्थी
यदि बा न होती तो शायद गांधी को यह ऊँचाई न मिलती (साक्षात्कार) 5 कथाकार गिरिराज किशोर से सत्येन्द्र शर्मा
सर–लेखन, भाव–पल्लवन साक्षात्कार प्रयोजन और कौशल
निर्धारित पाठ्य पुस्तक : ''हिन्दी भाषा संरचना'' मध्यप्रदेश हिन्दी ग्रन्थ अकादमी द्वारा प्रकाशित

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc
Semester – II

Course	Subject	Subject Code
B.Sc(Microbiology)	**Environment @ i;koj.k**	**FC-201/2**

Unit-I

Study of Environmental and ecology:

(a) Definition and Importance.

(b) Environmental Pollution and problems.

(c) Public participation and Public awareness.

पर्यावरण एवं पारिस्थितकीय अध्ययन

(क) परिभाषा एवं महत्व

(ख) पर्यावरण प्रदूषण एवं समस्याएँ

(ग) जनभागीदारी एवं जन जागरण

Unit-II

Environmental Pollution :

(a) Air, water, noise, heat and nuclear pollution.

(b) Causes, effect and prevention of pollution.

(c) Disaster management – Flood, Earthquake, cyclones and landslides.

पर्यावरणीय प्रदूषण

(क) वायु, जल, ध्वनि, ताप एवं आणविक–प्रदूषण

(ख) प्रदूषण के कारण, प्रभाव एवं रोकथाम

(ग) आपदा प्रबंधन– बाढ़, भूकंप, चक्रवात एवं भूस्खलन

Unit-III

Environment and social problems :

(a) Development – non-sustainable to Sustainable.

(b) Energy problems of cities.

(c) Water preservation – rain-water collection.

पर्यावरण और सामाजिक समस्याएँ

(क) अधारणीय से धारणीय विकास

(ख) नगरों की ऊर्जा समस्या

(ग) जल संरक्षण– वर्षा, जल–संग्रहण

Unit-IV

Role of mankind in conserving natural resources :

(a) Food resources – World food problem

(b) Energy resources – increasing demand for energy.

(c) Land resources – Land as resources.

प्राकृतिक संसाधनों के संरक्षण में मनुष्य की भूमिका

(क) खाद्य–आहार संसाधन – विश्व आहार समस्या

(ख) ऊर्जा संसाधन– ऊर्जा की बढ़ती मांग

(ग) भूमि संसाधन– भूमि संसाधन के गृप में, भूमि अवनयन, मनुष्यकृत भूस्खलन

Unit-V

Environment conservation laws :

(a) Conservation laws for air and water pollution.

(b) Wildlife conservation laws.

(c) Role of information technology in protecting environment & health.

पर्यावरण संरक्षण कानून

(क) वायु तथा जल प्रदूषण–संरक्षण कानून

(ख) वन्य प्राणि संरक्षण कानून

(ग) पर्यावरण तथा स्वास्थ्य रक्षा में सूचना प्रौद्योगिकी की भूमिका

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B.Sc.(Microbiology)

Syllabus
Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Microbiology)	Microbial Physiology & Microbial Genetics	BMB 121

Unit I

Cultivation and Pure Culture Techniques

Nutrition and nutritional types of bacteria, Bacteriological media (types and uses), cultivation of aerobic and anaerobic microbes, Isolation of microorganisms, pure culture and cultural characteristics.

Unit II

Microbial Growth

Mathematical expression of bacterial growth, generation time and growth rate, Growth curve and phases of growth cycle, Batch, continuous and synchronous cultures; diauxic growth, Factors affecting microbial growth.

Measurement and Preservation Methods

Quantitative measurement of bacterial growth by cell mass, cell number and cell activity, Maintenance and preservation of cultures,

Unit III

Control of Microorganisms- I

Microbial death curve under adverse condition, Concept of sterilization, disinfection, asepsis and sanitation, Physical methods of control- Temperature, radiation, desiccation, osmotic pressure, filtration.

Unit IV

Fundamentals of Genetics

DNA as genetic material, Structure and types of DNA and RNA, Genetic code, Protein synthesis - Transcription and translation.

DNA Replication and Gene Structure

DNA replication, Cis-trans complementation test, Fine structure analysis of r II region of T4 by Benzer.

Unit V

Mutation

Evidence for spontaneous nature of mutation, Molecular basis of mutation- Types of mutation, Types of bacterial mutants and their isolation, Mutagenic agents- Physical and chemical, Mutation rate and Ames test.

Genetic Recombination- I

Gene transfer in bacteria, Transformation- Competence, DNA uptake, artificially induced competence, electroporation, Transposable elements, Plasmid- Structure, properties and types of plasmids.

Reference Books:

1. Genes XI, Author- B. Lewin.
2. Principles of Genetics, Authors- Gardner, Simmons and Snustad.
3. Concepts of Genetics, Authors- Klug and Cummings.
4. Microbial Genetics, Authors- Freifelder.
5. Genetics, Authors- Arora and Sandhu.
6. Text of Microbiology, Authors- Ananthanarayanan and Paniker..
7. Textbook of Microbiology, Authors- Dubey and Maheshwari.
8. Microbiology, A Practical Approach. Authors- Patel and Phanse
9. Experiments in Biotechnology. Authors- Nighojkar and Nighojkar
10. General Microbiology, Authors- Powar and Daginawala.
11. Fundamentals in Microbiology, Authors- Frobisher and Hinsdinn.
12. Immunology, Microbiology and Biotechnology, Author- K.C. Soni.
13. Microbiology, Author- R.P.Singh.

List of Practicals:

1. Preparation of McFarland scale.
2. Use of counting chamber for bacterial count.
3. Effect of temperature on bacterial growth.
4. Effect of pH on bacterial growth.
5. Effect of osmotic pressure (salt and sugar concentration) on bacterial growth.
6. The oligodynamic action of heavy metals on bacterial growth.
7. One step growth of bacteriophage.

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Syllabus

Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	**Based on inorganic, organic and physical chemistry**	**BCH-121**

UNIT-I

Hydrogen Bonding – Definition, Types, effects of hydrogen bonding on properties of substances, application Brief discussion of various types of Vander Waals Forces

Metallic Bond and Semiconductors

Metallic Bond- Brief introduction to metallic bond, band theory of metallic bond Semiconductors- Introduction, types and applications.

UNIT-II

Rate of reaction, rate equation, factors influencing the rate of a reaction – concentration, temperature, pressure, solvent, light, catalyst. Order of a reaction, integrated rate expression for zero order, first order, second and third order reaction. Half life period of a reaction. Methods of determination of order of reaction.

Kinetics-II

Effect of temperature on the rate of reaction – Arrheniusequation. Theories of reaction rate – Simple collision theory forunimolecular and bimolecular collision. Transition state theory of Bimolecular reactions.

UNIT-III

Electrolytic conduction, factors affecting electrolytic ,.Arrhenius theory of ionization, Ostwald's Dilution Law. Debye-Huckel – Onsager's equation for strong electrolytes (elementary treatment only) Transport number, definition and determination by Hittorfs methods, (numerical included). **Electrochemistry-II** Kohlarausch's Law, calculation of molar ionic conductance and effect of viscosity. Application of Kohlarausch's Law in calculation of conductance of weak electrolytes at infinite diloution. Applications of conductivity, Definition of pH and pKa, Buffer solution, Buffer action, Henderson – Hazelequation, Buffer mechanism of buffer action.

UNIT-IV

Nomenclature of alkenes, , mechanisms of dehydration of alcohols and dehydrohalogenation of alkyl halides,. The Saytzeff rule, Hofmann elimination, physical properties and relative stabilities o falkenes. Chemical reactions of alkenes mechanisms involved in hydrogenation, electrophilic and free radical additions, Markownikoff's rule, hydroboration–oxidation, oxymercuration reduction, ozonolysis, hydration, hydroxylation and oxidation with $KMnO_4$.

UNIT-V

IUPAC nomenclature of branched and unbranched alkanes , thealkyl group, classification of carbon atoms in alkanes. Isomerismin alkanes, sources, methods of formation (with special reference to Wurtz reaction, Kolbe reaction Cycloalkanes nomenclature, synthesis of cycloalkanes and their derivatives –photochemical (2+2) cycloaddition reactions, dehalogenation of β,β-dihalides, ,pyrolysis of calcium or barium salts of dicarboxylic acids, Baeyer's strain theory and its limitations.

Book suggested :

1. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshow, Pergamon.
3. J. March., Advanced Organic Chemistry: Reactions, Mechanisms and Structure, John Wiley.
4. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
5. J. P. Lowe and K.Peterson, Quantum Chemistry Academic Press.

Paper (Practical's)
Section-A (Inorganic)
Paper Chromatography
Qualitative Analysis of the any one of the following Inorganic cautions and anions by paper chromatography (Pb2+ , Cu2+ , Ca2+ , Ni2+ , Cl- , Br- , I- and PO43-and NO3-

Section-B (Physical)
1. To determine the surface tension of a given liquid by drop number method.
2. To determine the viscosity of a given liquid**.**
3. To determine the specific refractivity of a given liquid.

SECTION – C (Organic)
1. Dibenzalacetone from acetone and benzaldehyde
2. Aspirin from salicylic acid. To study the process of) sublimation of camphor and phthalic acid

Books for practical:

1. Vogel's Textbook of Quantitative Analysis, revised, J. Bassett, R.C. Denney, G.H. Jeffery and J. Mendham, ELBS.
2. Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.
3. Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M.Miller, Prentice Hall.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Heal

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B.Sc.(Microbiology)

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Cell Biology & Genetics	BSB 121

Unit-1

The cell envelope: plasma membrane, bilayer lipid structure,function of the cell wall. Structure and function of cell organells: Golgibodies,ER, Peroxisome, Vacuole, Chloroplast and Mitochondrion.

Unit-2

Ultrastructure and function of nucleus: Nuclear membrane, Nucleolus, Extranuclear genome, Presence and functions of mitochondrial and plastid-DNA,Plasmids. chromosomal organization; morphology ,centromere and telomere, special types of chromosome, Mitosis and Meiosis

Unit-3

Variations in chromosomes structure : Deletions, duplications translocations. inversions; variation in chromosome number, aneuploidy, polyploidy, DNA the genetic material, DNA structure and replication, the nucleosome model, satellite and repetitive DNA.

Unit-4

Structure of gene: genetic code, transfer of genetic information; trascription, translation, protein syntesis, tRNA, and ribosomes. Regulation of gene expression in prokaryotes and eukaryotes.

Unit-5

Genetic inheritance: Mendelism; laws of segregation and independent assortment; linkage analysis; interactions of genes. Genetic variations; mutations, spontaneous and induced; transposable elements; DNA damage and repair.

Suggested Books :

1. Alberts B.D. Lewis, J.Raff, M.Rubers, K. and Watson I.D. 1999 molecular Biology of cell Garland Pub. Co. Inc. New York, U.S.A.
2. P.K. Gupta 1999 A text Book of cell and Molecular Biology, Rastogi Pub. Meerut India.
3. Kleinsmith L.J. and Molecular Biology (2nd edition) Harper Collins College pub. New York USA.
4. P.K. Gupta Genetic's Rastogi Pub. Meerut.
5. Sinha & Sinha cytogenetics & plant Breeding Vikas Pub.

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Syllabus

Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	CELL BIOLOGY & GENETICS	BSZ 121

UNIT 1

Prokaryotic and eukaryotic cells –Ultrastructure and Organization. Plasma membrane–Ultra structure–Chemical compositon and functions of modifications of plasma membrane. Endoplasmic reticulum: Morphology, Ultra structure, chemical composition and functions. Golgi complex: Ultra structure, chemical composition and functions.

UNIT 2

Lysosomes: Ultra structure and polymorphism- chemical composition and functions: Peroxisomes and glyoxysomes. Mitochondria: Ultra structure- chemical composition-enzyme systems- functions-Oxidation- Respiratory chain (ETP)- Kreb's cycle, ATP Production and Biogenesis.

UNIT 3

Ribosomes: Ultra structure-types- chemical composition - functions. Nucleus and Neucleolus: Ultra structure of Nucleus and Nucleolus. Nucleic Acids: DNA –Ultra structure-replication-transcription, Chromosomes: Ultra structure of Chromosomes and Giant Chromosomes, Cell division- mitosis and meiosis. Cell cycle,

UNIT 4

Introduction –Laws of Mendel –Interaction of genes (Epistatic gene ,Complementary genes and Lethal genes . Inheritance of Blood group in man and Coat colour in Rabbit .
Mechanism of linkage and crossing over –Types and theories - Chromosomal
Mapping –Sex linked inheritance (haemophilia, and colour blindness). Sex limited inheritance and sex influenced inheritance .

UNIT 5

Sex determination in man, Drosophila and Bonellia. Mutations – Point mutation and Chromosomal aberrations and mutagens **.** Inbreeding and out breeding, heterosis –Genetic applications in animals. DNA as genetic material –Experiments . Syndromes (Down syndrome and Turners syndrome in man) **.**

TEXT BOOKS:

Cell biology. Veer Bala Rastogi, Rastogi Publications.
Cell Biology, Power.Verma P.S. and Agarwal V. K . –Concepts of Genetics .
Rastogi V.B. A text book of Genetics, K.Ramnath, Meerut.

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YEAR-I,SEMESTER-II
PRACTICAL
CELL BIOLOGY AND GENETICS

A. Major Practicals

Use Microscopes, Camera Lucida, Stage and Ocular micrometers.

Total Counting of RBC / WBC Using haemocytometer.

Blood Smear Prepartion, Differential count of WBC.

Mounting Buccal Epithelium and observing living cells using vital staining.

Study of mitotic division using onion root tips.

Study of prepared slides of different tissues.

Submission of practical record.

B.GENETICS PRACTICALS
Observation of common mutants of drosophila
Preparation of mounting of the salivary gland in chironomous larva
Submission of practical record.

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SCHEME

Semester – III

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-301/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-301/2	Core	Environment	10	4	40	14	-	-	50
3	BMB 131	Core	Microbial Biochemistry & Environmental Microbiology	20	8	80	27	50	17	150
4	BCH 131	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 131	Optional	Diversity & systematics of seed plants	20	8	80	27	50	17	150
	BSZ 131	Optional	Animal Physiology & Developmental Biology And Immunology	20	8	80	27	50	17	150
			Total	**80**	**32**	**320**	**109**	**150**	**51**	**550**

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Semester – III

Course	Subject	Subject Code
B.Sc(Microbiology)	**Values & Spirituality**	**FC-301/1**

Chapter 1: VALUE EDUCATION

1.1	Objectives	1.2	Introduction	1.3	Concepts of Values
1.4	Definition and Types of values			1.5	The need for Education in values
1.6	Challenges for Value adoption			1.7	Character development
1.8	Vision of a better world			1.9	Summary
1.10	Glossary			1. 11	Suggested reading

Chapter 2: INCULCATION OF VALUES

2.1	Objectives	2.2	Introduction	2.3	Classification of values
2.4	Personal Values	2.5	Family Values	2.6	Social Values
2.7	Spiritual values	2.8	Benefits of value adoption	2.9	Summary
2.10	Glossary	2.11	Suggested reading		

Chapter 3: MAJOR RELIGIONS OF THE WORLD

3.1	Objectives	3.2	Introduction	3.3	Hinduism
3.4	Jainism	3.5	Buddhism	3.6	Christianity
3.7	Islam	3.8	Sikhism	3.9	Summary
3.10	Glossary	3.11	Suggested reading		

Chapter 4: EXPLORING THE SELF

4.1	Objectives	4.2	Introduction	4.3	Anatomy of the self
4.4	The cyclic processes within the self			4.5	States of the awareness
4.6	Innate qualities	4.7	Acquired qualities	4.8	Empowering the self
4.9	Summary	4.10	Glossary	4.11	Suggested reading

Chapter 5: THOUGHT AND THE THINKER

5.1	Objectives	5.2	Introduction	5.3	Know the mind(TEAM)
5.4	Thought power	5.5	Types of thoughts	5.6	Thinking process
5.7	Positive thinking	5.8	Power and Acts	5.9	Summary
5.10	Glossary	5.11	Suggested reading		

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Semester – II

Course	Subject	Subject Code
B.Sc(Microbiology)	Environment	FC-301/2

Unit - I

Problems of natural resources:

(a) Problems of water resources- Utilization of surface and ground water overutilization, flood, draught, conflict over water, Dams and related problems.

(b) Problems of forest resources -uses and over utilization, deforestation, Dams and its effects on forests and tribes.

(c) Problems of land resources -land as a source , erosion of land, man-induced, land slides and desertification.

प्राकृतिक संसाधन की समस्याएँ

(क) जल संसाधन की समस्या– सतह एवं भूजल का उपयोग , अति दोहन बाढ़ सूखा , जल पर संघर्ष , बॉध –लाभ एवं समस्याए

(ख) वन संसाधन की संस्याएँ –उपयोग एवं अति दोहन , वनोन्मूलन , इमारती लकड़ी, निस्सारण , बाधँ एवं उनका बन और आदिवासीयो पर प्रभाव

(ग) भूमि संसाधन की समस्याएँ– स्त्रोत के क्रय में भूमि का अवभ्रमण , मानव प्रेरित भू – स्खलन और मरूस्थीकरण

Unit - II

Bio-diversity and its protection -

(a) Value of bio-diversity -consunable use: productive use, social, alternative, moral asthetic and values

(b) Bio - diversity and multi -diversity at global and national levels.

(c) Threats to bio -diversity -loss of habitat, poaching of widlife, man wildlife conflicts.

जैव विवधिता और उसका संरक्षण –

(क) जैव विविधता का मूल्य – उपभोग्य , उपयोग , उत्पादक उपयोग, सामाजिक, नैतिक सौर्दयंगत तथा वैकल्पिक मूल्य

(ख) वैश्विक , राष्ट्रीय तथा स्थानीय स्तरो पर जैव विविधता वृहत विविधताओ के राष्ट्र रूप में भारत ।

(ग) जैव विविधता के खतरे – आवासीय हानि ,वन्य जीवन में अनाधिकार घुसपैठ तथा मानव , वन्य जीवन –संघर्ष

Unit - III

Human population and environment

(a) Population growth, disparities between countries.

(b) Population explosion, family welfare programme

(c) Environment and human health

पर्यावरण और जनसंख्या तथा पर्यावरण

(क) जनसंख्या – वृद्धि , राष्ट्रो के बीच अंतर

(ख) जनसंख्या –विस्फोट , परिवार कल्याण कार्यक्रम

(ग) पर्यावरण और मानव स्वास्थ्य

Unit - IV

Multidisciplinary nature of environmental studies :

(a) Natural resources

(b) Social problems and the environment

(c) Environmentel awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

(क) प्राकृतिक संसाधन

(ख) सामाजिक समस्याएँ और पर्यावरण

(ग) पारिस्थिति तन्त्र

Unit - V

Multidisciplinary nature of environmental studies :

(a) Natural resources

(b) Social problems and the environment

(c) Environmentel awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

(क) प्राकृतिक संसाधन

(ख) सामाजिक समस्याएँ और पर्यावरण

(ग) पारिस्थिति तन्त्र

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B.Sc.(Microbiology)

Syllabus

Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Microbiology)	Microbial Biochemistry & Environmental Microbiology	BMB 131

Unit I

Carbohydrates

Chemical structures, nature and properties, Classification and importance in biological cells, Aerobic and anaerobic metabolism. Amino acids- Classification and properties. Structure, Zwitterion nature, Proteins- Classification, Structure and function. Primary, secondary, tertiary and quaternary structure, Proteolysis,

Unit II

Enzymes & Bioenergetics

General characteristics. Factors affecting enzyme activity, Regulation of enzyme activity, Enzyme kinetics, Km, activation and inhibition, Coenzymes and cofactors. Non-protein enzymes Applications of enzymes. Principles of bioenergetics and high energy phosphate compounds, Mode of energy production- Photophosphorylation, Bacterial photosynthesis.

Unit III

Lipids, vitamins and hormones

Saturated and unsaturated fatty acids, Structure, classification, properties and function of lipids and vitamins, Distribution and functions of lipids in microorganisms, Beta-oxidation of lipids, Hormones: Steroid hormones, Structure and function.

Unit IV

Soil Microbiology, Water Microbiology

Formation and composition of soil, Estimation of soil microflora, Soil management, Rhizosphere- Positive and negative interactions among soil microflora. Microbiology of water and water bodies, Water purification, Eutrophication.Primary treatment, Secondary treatment, Advanced and final treatment.

Unit V

Food Microbiology & Air Microbiology

Introduction to microbiology of food and milk, Food intoxications, spoilage of food- Fresh food, canned food, vegetables and milk products, Preservation of food and milk, Composition of milk, grading of milk- MBRT,. Composition and analysis of air, Aeromicroflora of different habitats, Aeroallergens, Biogeochemical cycles- Role of microbes in Nitrogen and Carbon cycles.

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SEMESTER-III

Recommended Books (Semester-III)

1. Principles of Biochemistry, Author- A.L. Lehniger
2. Fundamentals of Biochemistry, Author- J. L. Jain
3. Biochemistry, Author- Voet and Voet.
4. Textbook of Biochemistry- S.P. Singh.
5. Biochemistry, Author- Stryer.
6. Introduction to protein structure, Authors- Branden and Tooze.
7. Fundamental Principles of Bacteriology, Author- A.J. Salle.
8. Principles of Biochemistry, Authors – Zubey, Parson and Vance.
9. Microbial Diversity, Author- D. Colwd.
10. Microbiology A Practical Approach Authors- Patel and Phanse, .
11. Nighojkar and Nighojkar, Experiments in Biotechnology.
12. Food Microbiology, Authors- Frazier and Westhoff.
13. Food Microbiology, Authors- Adams and Moss
14. Introductory Food Microbiology. Author – H.A. Modi
15. Environmental Microbiology, Author- P.D. Sharma.
16. Environmental Microbiology, Author- K.G. Vijaya.
17. The nature and properties of soil. Authors- Harry buckman and Nyle C. brady.
18. Introduction to soil Microbiology Internationals. Authors- Martin Alexander.

YEAR-I SEMESTER-III

LIST OF EXPERIMENTS

1. Detection of carbohydrates, proteins and lipids.
2. Estimation of activity of enzymes like amylase, protease and lipase.
3. Effect of pH on enzyme activity.
4. Effect of temperature on enzyme activity.
5. Effect of substrate concentration on enzyme activity.
6. Effect of enzyme concentration on enzyme activity.
7. Quantitative estimation of protein by Folin Lowry's Mehod.
8. Quantitative estimation of carbohydrates by Nelson Smogyi's Method.
9. Isolation of organisms from air.
10. Isolation of organisms from water and sewage.
11. Isolation of organisms from food sources.
12. Isolation of Yeast.
13. Isolation of phosphorous solubilizing bacteria/fungus from soil sample.
14. Isolation of *Xanthomonas citri* from citrus canker.
15. Gradation of milk by Methylene Blue Reduction Test (MBRT).

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Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Based on inorganic, and Physical Chemistry	BCH-131

UNIT-I

Definition of transition elements, position in the periodic table, General characteristics & properties of Ist transition elements, Structures & properties of some compounds of transition elements– TiO_2, $VOCl_2$, $FeCl_3$, $CuCl_2$ and Ni $(CO)_4$.**Chemistry of Elements of IInd & IIIrd transition series** General characteristics and properties of the II^{nd} and III^{rd} transition elements Comparison of properties of 3d elements with4d & 5d elements with reference only to ionic radii, oxidation state, magnetic and Spectral properties and stereochemistry.

UNIT-II

Werner's coordination theory, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, valence bond theory of transition metal complexes. **Non-aqueous Solvents** Physical properties of a solvent, types of solvents and their general characteristics, reactions in non-aqueous solvents with reference to liquid NH_3 and liquid SO_2.

UNIT-III

Definition of thermodynamic terms: system, surrounding etc. First law of thermodynamics: statement, definition of internal energy and enthalpy. Heat capacity, heat capacities at constant volume and pressure and their relationship. Joule’s law – Joule – Thomson coefficient for ideal gas and real gas: and inversion temperature. **Thermodynamics-II** Calculation of w.q. dU & dH for the expansion of ideal gases under isothermal and adiabatic conditions for reversible process, Temperature dependence of enthalpy, Kirchoffs equation.Bond energies and applications o f bond energies.

UNIT-IV

Monohydric alcohols nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding. Acidic nature.Reactions of alcohols. **Epoxides** Synthesis of epoxides. Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening.

UNIT-V

Nomenclature of Carboxylic acids, structure and bonding, physical properties, acidity of carboxylic acids,. Preparation of carboxylic acids. Reactions of carboxylic acids. **Acid Derivatives** Structure , nomenclature and preparation of acid chlorides, esters, amides and acid anhydrides. Relative stability o f acyl derivatives. Physical properties, inter convers ion of acid derivatives by nucleophilic acyl substitution. Mechanisms of esterification and hydrolysis (acidic and basic).

Book suggested :

1. R. G. Mortimer, Mathematics for Physical Chemistry Elsevier.
2. F. L. Pilar, Elementary Quantum Chemistry , Dover Publication
3. E. L. Eliel, Stereochemistry of Carbon Compounds, McGraw-Hill
4. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry,
5. Comprehensive Coordiantion Chemistry eds., G. Wilkinson, R.D. Gillars and J.A. Mc Cleverty,Pergamon.

Paper (Practicals)

Section – I (Inorganic)

1**. Gravimetric Analysis**

Quantitative estimations of, Cu^{2+} as copper thiocyanate and Ni^{2+} as Ni – dimethylglyoxime.

Section-B (Physical)

1. To determine the CST of phenol – water system.
2. To determine the solubility of benzoic acid at various temperatures and to determine the H of the dissolution process
3. To determine the enthalpy of neutralisation of a weak acid/weak base vs. strong base/strong acid and determine the enthalpy of ionisation of the weak acid/weak base.

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: Naphthalene , anthracene, acenaphthene, benzyl chloride, *p*-dichlorobenzene, *m*-dinitrobenzene, *p*-nitrotoluene, resorcinol , hydroquinone, □ -naphthol, □ -naphthol, benzophenone, ethyl methyl ketone, benzaldehyde, vanillin, oxalic acid, succinic acid,

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Third Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Diversity & systematics of seed plants	BSB 131

UNIT – 1

Characteristics and Classification of Gymnosperms, Heterospory and Origin of Seed Habit, Evolution and Diversity of Gymnosperms, Geological Time Scale, and Fossilization.Fossil Gymnosperms: Lyginopteris and Lagenostoma.

UNIT – 2

Morphology, Anatomy Reproduction and life cycle of Cycas, Pinus and Ephedra.

UNIT – 3

Origin and Evolution of Angiosperms, Fundamental components of α, β, γ taxonomy, Plant Identification, Principles and rules of Botanical Nomenclature, Herbarium and Botanical gardens; Classification of Angiosperms: Bentham and Hooker, and Hutchinson, Modern trends in Taxonomy.

UNIT –4

Diagnostic characteristics and Economic Importance of Families –Ranunculaceae, Brassicaceae, Malvaceae, Rutaceae, Fabaceae, and Apiaceae.

UNIT – 5

Diagnostic characteristics & Economic Importance of Families –Asteraceae, Asclepiadaceae, Solanaceae, Lamiaceae, Euphorbiaceae, Liliaceae and Poaceae.

Practical Exercises + Scheme (Marks- 50)

Gymnosperms- 10

-Morphological and anatomical study of Cycas, Pinus, and Ephedra (all parts).

- Study of permanent slides of Cycus, Pinus and Ephedra.

Angiosperms- 15

-Study of types of inflorescence and flowers with labelled sketches.

- Technical description of common flowering plants belonging to families mentioned in theory syllabus.

Spotting- 10

Viva- voce- 5

Practical record - 10

SUGGESTED READINGS:--

1. Agarwal, S.B. 2007. Unified Botany, Shivlal Agarwal & Company Indore.

2. Bhatnagar, S. P. and Moitra 1996. Gymnosperms. New Age International Limited, New Delhi.

3. Davis, P.H. and Heywood, V.H. 1963, Principles of Angiosperm taxonomy. Oliver and Boyd, London.

4. Gangulee, H. C. & Kar, A. K. 2006. College Botany Voll.III, New Central Book Agency (P) Ltd. Kolkata, 700009.

5. Heywood, V.H. and Moore, D.M. (eds) 1984. Current concepts in plant taxonomy. Academic press London.

6. Jeffery, C. 1982. An Introduction of plant taxonomy. Cambridge University Press Cambridge, London.

7. Jones, S.B. Jr. and Luchsinger, A.E. 1986. Plant Systematic. Mc Graw Hill Book Co. New York.

8. Kaushik, M.P. 2003. Modern Textbook of Botany, Prakash Publication Muzaffar Nagar U.P.

9. Mukherjee, S.K. 2006. College Botany Voll.II, New Central Book Agency (P) Ltd. Kolkata, 700009.

10. Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.

11. Radford, A.E. 1986. Fundamentals of Plant Systmatics, Happer and Raw, New York.

12. Saxena and Sarabhai. 1989. Text book of Botany. Rastogi Publication Meerut.

13. Singh, G. 1999. Plant Systematics : Theory and Practice. Oxford and IBH Pvt. Ltd. New Delhi.

14.Vasishta, P.C. 2005. Botany for degree students Voll. V, Gymnosperms. S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus

Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	ANIMAL PHYSIOLOGY & DEVELOPMENTAL BIOLOGY AND IMMUNOLOGY	BSZ 131

Unit-1

Nutrition –types, Enzymes – Enzyme action, Coenzymes, Digestion in man. Respiration – Respiratory pigments, role in transport of O2 and CO2 in man. Circulation - blood composition, origin and conduction of heart beat in an – blood pressure, Heart diseases– heart attack.

Unit-2

Excretion – types of nitrogenous wastes – structure of the mammalian kidney and urine formation – renal failure – kidney stone – kidney transplantation. Osmotic – ionic regulation in freshwater, marine, estuarine and terrestrial organisms

Unit-3

Amoeboid, ciliary and flagellar movements. Types of muscles – ultra structure of skeletal muscle – Muscle contraction and theories. Neuron, - Types- Impulse transmission, -synaptic transmission - reflex action.

-Endocrine glands in man, secretions and disorders.

Unit-4

Spermatogenesis – definition – process and significance, structure of mammalian sperm. Oogenesis – definition – process and significance – Types of eggs and egg membranes. Fertilization – definition – process and significance. Parthenogenesis – definition and significance – types of parthenogenesis.

Unit-5

Cleavage patterns (types) – Cleavage in Frog, Chick and Mammals. Morula and Blastulation.

Introduction – cells and organs involved in immune response. Types of immunity – Innate & adaptive immunity of acquired immunity, humoral and cell mediated immunity, active and passive immunity.

TEXT BOOKS:

Verma P.S. & Tyagi B.S. Animal Physiology, 6th edition. S.Chand & Co.
Agarwal, V.K. Agarwal, R.A.Srivastava A.K. & Kausha Kumar, Animal physiology & Biochemistry, S. Chand & Co.,
De Beer, G.R. Embryos and Ancestors. Clarenden Press, Oxford.
Verma. P.S and Agarwal, V.K. Chordate Embryology, S.Chand and Co. Ltd., New Delhi (1998).
Bodmer, Modern Embryology, Saunders International student edition, Philadelphia.3 rd Edition 1981**.**
Eli Benjamini et al.,(1991) Immunology – A short course – Wiley Publishers, NY.

REFERENCES:
Hoar, W.S (1987) General and Comparative physiology, prentice – Hall.
M.K.Chanddrashekaran – Circadian Rhythms – Madras science foundation, Chennai**.**

YEAR-II SEMESTER-III

PRACTICAL

(ANIMAL PHYSIOLOGY, AND DEVELOPMENTAL BIOLOGY)

I Major Practicals:

1) Qualitative analysis of digestive enzymes in cockroach.

2) Estimation of the rate of o2 consumption in fish/crab with reference to body weight.

3) Detection nitrogenous waste products in fish tank water, bird excreta & mammalian urine.

4) Study of human salivary activity in relation to temperature.

5) Qualitative analysis of carbohydrates, proteins, and amino acids.

II Minor Practicals:

Kymograph –simple twitch, Trappe, Fatigue, Tetanus, Spigmomanometer, pH meter, Colorimeter, Haemometer, Enzyme action – graphs (temperature, concentration of substrate and enzyme.)

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

SCHEME

Semester – IV

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-401/1	Core	English Language	10	4	40	14	-	-	50
2	FC-401/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BMB 141	Core	Immunology & Medical Microbiology	20	8	80	27	50	17	150
4	BCH 141	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 141	Optional	Structure, development & reproduction in flowering plants	20	8	80	27	50	17	150
	BSZ 141	Optional	Ecology & Evolutionary Biology	20	8	80	27	50	17	150
			Total	80	32	320	109	150	51	550

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B.Sc.(Microbiology)

B.C.A./B.A./B.Com./B.Sc
Semester – IV

Course	Subject	Subject Code
B.Sc(Microbiology)	**English Language**	**FC-401/1**

Unit-I

1.	William Wordsworth	:	"The World is Too Much With Us"
2.	K. Aludiapillai	:	"Communication Education and Information Technology" "Democratic Decentralisation"
3.	S. C. Dubey	:	"Basic Quality of Life"
4.	Sister Nivedita	:	"The Judgment Seat of Vikramaditya"
5.	Juliun Huxley	:	"War as a Biological Phenomenon"
6.	Robert Frost	:	"Stopping by Woods on a Snowy Evening"
7.	Ruskin Bond	:	"The Cherry Tree"

Unit-II

Short Essay of about 250-300 words.

Unit-III

Translation of a short passage from Hindi to English.

Unit-IV

Drafting CV, writing e-mail message for official purpose.

Unit-V

Language Skills, One-word substitution, homonyms, homophones, words that confuse, Punctuation, Idioms

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./B.Com./B.Sc

Semester – IV

Course	Subject	Subject Code
B.Sc(Microbiology)	**Development of Entrepreneurship**	**FC-401/2**

Unit-I

Entrepreneurship - Meaning, Concept, Characteristics of entrepreneur.

उद्यमिता – का आशय, मत, उद्यमिता के गुण।

Unit-II

Types of entrepreneurship, importance and views of various thinkers (Scholars).

- Formation of goals, How to achieve goals.
- Problems in achieving targets and solution.
- Self motivation, elements of self motivation and development.
- Views of various scholars, evaluation, solutions.

Leadership capacity : Its development and results.

उद्यमिता के प्रकार, महत्व और विभिन्न विद्वानों के मत, लक्ष्य निर्माण, लक्ष्य कैसे प्राप्त करें। लक्ष्य प्राप्ति में समस्याऍं, उनका समाधान, स्वप्रेरणा, स्वप्रेरणा के तत्व और विकास, विभिन्न विद्वानों के मत, आकलन, निष्कर्ष , नेतृत्व समता, उसका निकास और प्रतिफलन

Unit-III

Projects and various organisations (Govt., non-Govt.), Govt. Projects, Non-Govt. projects. Contribution of Books, their limitations, scope.

परियोजनाऍं तथा विभिन्न संगठन (शासकीय–अशासकीय), शासकीय परियोजनाऍं:– अशासकीय परियोजनाऍं:– बैंको का योगदान, उनकी सीमाऍं, क्षेत्र

Unit-IV

Functions, qualities, management of a good entrepreneur. Qualities of the entrepreneur (Modern and traditional). Management skills of the entrepreneur. Motive factors of the entrepreneur.

अच्छे उद्यमी के कौन–कौनसे कार्य, गुण, प्रबंधन इत्यादि, अच्छे उद्यमी के गुण आधुनिक और पूर्ववर्ती, उद्यमी की प्रबंधन कला, उद्यमी के प्रेरक तत्व

Unit-V

Problems and Scope of the Entrepreneur :

- Problem of Capital
- Problem of Power
- Problem of registration
- Administrative problems
- Problems of Ownership.

उद्यमी की समस्याऍं, क्षेत्र, पूॅंजी की समस्या, शक्तिकरण की समस्या, पूॅंजीवन की समस्या, प्रशासनिक समस्याऍं, स्वामित्व की समस्याऍं, इत्यादि

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Microbiology)	IMMUNOLOGY & MEDICAL MICROBIOLOGY	BMB 141

Unit I

Infection
Normal flora of human body, Infection and its types, Mechanism of pathogenesis.

Immune System
Organs of Immune system- Spleen, thymus and lymph nodes, Cells of Immune system- T cells- its types and receptors. B cells and its receptors.

Unit II

Immune Response
Immunity- Innate and acquired, Host defense mechanism- First, second and third line of host defense, Primary and secondary responses.

Antigens and Antibodies
Antigens- Properties and types, Adjuvants, Immunoglobulins- Separation, structure and types, Generation of antibodies, Antibody diversity.

Unit III

Antigen and Antibody Reactions
Agglutination and precipitation reactions, Hemagglutination and PHA, Immunofluorescence, ELISA, RIA, Coombs test (Direct and Indirect), Complement- Components and biological activities.

Unit IV

Epidemiology of Infectious Diseases
Epidemiological study, Transmission of diseases, Types of diseases- Epidemic, pandemic and sporadic, Nosocomial infections.

Antimicrobial Agents
Antibiotics- Mode of action, Development of resistance, Transmission of drug resistance, ntiviral and antifungal drugs.

Unit V

Hypersensitivity
Hypersensitivity- Immediate and delayed type, Autoimmune diseases, Skin tests.

Microbial Diseases- I
Gram Positive Cocci- *Staphylococcus aureus* and *Streptococcus pneumonia*, Gram Negative Bacilli- *Salmonella typhi* and *Vibrio cholarie*, Acid fast bacteria- *Mycobacterium tuberculosis.*
Virus- Hepatitis and HIV.

Recommended Books (Semester-IV)

1. Immunology, Author- J. Kuby.
2. Fundamental Immunology, Author– W.E. Paul.
3. Fundamentals of Immunology, Authors– Coleman, Lombord and Sicard.
4. Immunology – Weir and Steward.
5. Immunology, A Textbook, Author- C.V. Rao.
6. Lecture Notes in Immunology, Author- I.R.Todd.
7. Essentials of Immunology, Authors- Roitt, I.M.
8. Immunology-Understanding of Immune System, Author- Klaus D. Elgert (1996)
9. Text Book on Principles of Bacteriology, Virology and Immunology, Authors- Topley &
10. Wilson's (1995)
11. The Experimental Foundations of Modern Immunology. Author- Clark, V.R.,
12. Medical Microbiology, Vol. 1 : Microbial Infection, Vol. 2 : Practical Medical Microbiology,
13. Authors- Mackie and McCartney.
14. Epidemiology and Infections, Author- Smith
15. Lecture Notes in Immunology, Author- I.R. Todd
16. Microbiology in Clinical Practice, Author- D.C. Shanson.
17. Diagnostic Microbiology, Authors- Baron, Peterson and Finegold.

YEAR-II SEMESTER-IV

LIST OF EXPERIMENTS

1. Estimation of haemoglobin by Sahli's method.
2. Estimation of haemoglobin by Cyname haemoglobin mehod.
3. Total count of W.B.C.
4. Total count of R.B.C.
5. Differential W.B.C. count.
6. Flocculation reaction- VDRL
7. Agglutination reaction- Widal test, Blood Grouping.
8. Immuno-diffusion techniques- ODD and RID.
9. UV as a mutagenic agent.
10. Replica plating technique.
11. Estimation of skin microflora.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	BASED ON INORGANIC, AND PHYSICAL CHEMISTRY	BCH-141

UNIT-I

Electronic structure, oxidation states and ionic radii and lanthanide contraction, complex formation, occurrence and isolation, lanthanide compounds. **Actinides-**General features and chemistry of actinides, chemistry of separation of Np, Pu and Am from U, Comparison of properties of Lanthanides and Actinides and with transition elements .

UNIT-II

Second law of thermodynamics, need for the law, , Carnot's cycles and its efficiency, Carnot's theorem, .Thermodynamics scale of temperature. Concept of entropy - entropy as a state function, entropy as a function of V & T, entropy as a function of P & T,

Thermodynamics-IV Third law of thermodynamics: Nernst heat theorem, statement of concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function(G) and Helmholtz function (A) as thermodynamic quantities, A &G as criteria for thermodynamic equilibrium and spontaneity, their advantage over entropy change. Variation of G and A with P, V and T.

UNIT-III

Electrolytic and Galvanic cells – reversible & Irreversible cells ,conventional representation of electrochemical cells. EMF of cell and its measurement, Wes ton standard cell, activity and activity coefficients .Calculation of thermodynamic quantities of cell reaction (ΔG, ΔH & K). Types of reversible electrodes – metal- metal ion gas electrode, metal –insoluble salt-anion and redox electrodes. Electrode reactions, Nernst equations, derivation of cell EMF and single electrode potential. Standard Hydrogen electrode, reference electrodes, standard electrodes potential, sign conventions electrochemical series and its applications.

UNIT-IV

Structure and nomenclature of amines, physical properties. Separation of a mixture of primary, secondary and tertiary amines. Preparation of alkyl and aryl amines (reduction of nitro compounds, nitriles, reductive amination of aldehydic and ketonic compounds. Gabrielphthalimide reaction, **Diazonium Salts** Mechanism of diazotisation, structure of benzene diazoniumchloride, Replacement of diazo group by H, OH, F, Cl, Br, I, NO2reaction and its synthetic application.

Nitro Compounds Preparation of nitro alkanes and nitro arenes and their chemical reactions.

UNIT-V

Nomenclature and structure of the carbonyl group. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chlorides, **Ketones** Mechanism of nucleophilic additions to carbonyl group with particular emphasis on benzoin, aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives. Wittig reaction. Mannichreaction. Oxidation of aldehydes, Baeyer–Villiger oxidation of ketones, Cannizzaro reaction. MPV, Clemmensen, Wolff-Kishner, $LiAlH_4$ and $NaBH_4$ reductions.

Book suggested :

1. I. L. Levine, Quantum Chemistry , Prentice-Hall Inc., New Jersey.
2. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
3. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry, Macmillan
4. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.

(Practicals)

SECTION – I (Inorganic)

Colorimetry:

1. To verify Beer - Lambert law for $KMnO_4/K_2Cr_2O_7$ and determine the concentration of the given $KMnO_4/K_2Cr_2O_7$ solution.
2. Preparations: Preparation of Cuprous chloride, prussion blue from iron fillings, tetraammine cupric sulphate, chrome alum, potassium trioxalatochromate (III).

Section-B (Physical)

3. To determine the enthalpy of solution of solid calcium chloride
4 .To study the distribution of iodine between water and CCl_4.

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: benzoic acid,salicyclic acid, aspirin, phthalic acid, cinnamic acid, benzamide, urea, acetanilide, benzanilide, aniline hydrochloride, p-toluidine, phenyl salicylate (salol), glucose, fructose, sucrose, *o-, m-, p*nitroanilines,thiourea.

Books for practical:

1. Vogel's Textbook of Practical Organic Chemistry, A.R. Tatchell, John Wiley.
2. Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.
3. Findley's Practical Physical chemistry, B.P. Levitt, Longman.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Health.

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B.Sc.(Microbiology)

Fourth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	**Structure, development & reproduction in flowering plants**	**BSB 141**

UNIT –1

The Root system: Root apical meristems, Differentiation of primary and secondary tissues and their roles, Anatomy of Monocot and Dicot roots, Morphological modification of root for storage, respiration, reproduction and interaction with microbes.

UNIT – 2

The Shoot system: Shoot apical meristem and histological organization, Anatomy of primary stem in Monocotyledons and Dicotyledons, Secondary growth in stem and root – Vascular cambium and its functions, Characteristics of growth rings, Sapwood and Heart wood, Secondary Phloem, Cork Cambium and Periderm.

UNIT –3

The Leaf system: Origin, Development, Diversity in size, shape and arrangement, Internal structure of Dicot and Monocot leaf in relation to photosynthesis and water loss, Adaptations to water stress, senescence and abscission.

UNIT – 4

The Flower system: Concept of flower as a modified shoot, Structure of Anther, Microsporogenesis and Male Gametophyte, Structure of Pistil, Ovules, Megasporogenesis and Development of Female Gametophyte (Embryo Sac) and its types, Pollination – Mechanism and Agencies of Pollination, Pollen Pistil interactions and Self incompatibility.

UNIT – 5

Double Fertilization, Development and types of Endosperm and its morphological nature, Development of Embryo in Monocots and Dicots, Fruit development and maturation. Seed structure and dispersal, Vegetative Propagation.

Practical Exercises + Scheme

(Marks- 50)

1- Cutting, staining and mounting of cross section of two materials of monocotyledons/dicotyledons root and stem and leaf like Sunflower and Maize or other available material. 15

2- Organisation of shoot Apex and Root Apex. 5

3- Study of Ovules and Anthers and their types 5

- Structure of stigma and style (Hibiscus, Maize, Ocimum, Citrus and Clitoria

(Aprajita) or plant studied by you.

4-Spotting- 10

5-Viva- voce- 5

6-Practical Record- 10

SUGGESTED READINGS:--

1•Gangulee, H.C., Das, K. S. And Dutta, C. 2007. College Botany Voll.I, New Central Book Agency (P) Ltd. Kolkata, 700009.

2•Hywood, V.H. & Moore, D.M. (eds) 1984. Current concepts in plant taxonomy. Acedemic press London.

3•Jones, S.B. Jr. and Luchsinger, A.E. 1986, Plant taxonomy (III edition) Mc Graw Hill Book Co. New York.

4•Maheshwari, P.1978. Plant Embryology.

5• Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.

6• Radford, A.E. 1986. Fundamentals of Plant Systematics, Harper and Row, New York.

7• Shrivastava and Das. Modern text book of Botany Vol-III & IV.

8• Singh, V., Pande P.C. and Jain , D. K. Structure & Development in Angiosperms. Rastogi Publication, Meerut.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	ECOLOGY & EVOLUTIONARY BIOLOGY	BSZ 141

UNIT I

Abiotic factors of the Environment : Temperature, Light and .Oxygen. Bio geo chemical cycles with special reference to Nitrogen Phosphorous and Carbon. Biotic factors of the environment - Animal relationship.

UNIT II

Population : characteristics –Natality, Mortality, Density, and age distribution, population control, life-tables, Food chains, Food webs and Ecological pyramids. Air pollution, Water pollution and Oil pollution. Noise pollution and Thermal pollution.

UNIT III

Pond as a Ecosystem, energy flow and ecological succession. Habitats – Terrestrial – Aquatic – Marine, Fresh water and estuary. Environmental resources- renewable and non renewable resources. Forest resources- Protection – Chipko movement- A forestation. Wild life management- Wild life sanctuaries and National Parks.

Unit-IV

History of Evolutionary thought - Origin of life –Chemical evolution. Evolution of self replicating systems –DNA world and RNA world.

Evidences from Paleontology -Comparative anatomy, Embryology, Physiology and Bio chemistry. Bio geography –Distribution in continents ,Continuous and discontinuous distribution - Endemism .

Unit-V

Natural selection- Species and Speciation- Sympatric and allopatric speciation. Isolating mechanism-mutation and genetic drift.

Adaptation and adaptive radiation, Colouration-mimicry-Darwins finches. Polymorphism-types and significance. Convergent-Divergent-parallel and co-evolution of Man and cultural evolution.

TEXT BOOKS

1. H.D.Kumar, Modern concepts of Ecology. Vikas Publishing house.
2. E.P. Odum, Fundamentals of Ecology.
3. G.C. Clarke, Elements of Ecology, John Wiley sons, New York
4. Rostogi, V.B. Organic Evolution, Kedernath, Ramnath publishers, Meerut.

5, Verma P.S. & Agarval, V.L. concepts of evolution S.Chand& Company.

REFERENCES:

Introduction to evolution-Dodson-Evolution: process and product.

PRACTICAL

Spotters:

Description and uses of autoclave, Hot air oven, Incubator, Water both, Centrifuge, Refrigerator, pH meter, Colori meter, Microtome, Rain gauge, Anemometer, Maximum minimum thermometer, Hygrometer, and Barometer.

Computer applications - Hardware of computer, storage device, mouse.

Submission of field Report.

Submission of Practical Record.

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B.Sc.(Microbiology)

SCHEME

Semester – V

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-501/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-501/2	Core	Basic Computer Information Technology-I	10	4	40	14	-	-	50
3	BMB 151	Core	Bioinformatics , Biostatistics & Industrial Microbiology	20	8	80	27	50	17	150
4	BCH 151	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 151	Optional	Plant Physiology And Biochemistry	20	8	80	27	50	17	150
	BSZ 151	Optional	Microbiology And Biochemistry	20	8	80	27	50	17	150
			Total	**80**	**32**	**320**	**109**	**150**	**51**	**550**

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc
Semester – V

Course	Subject	Subject Code
B.Sc(Microbiology)	**Values & Spirituality**	**FC-501/1**

Chapter 6: THE SUPREME- SOURCE OF VALUES

6.1 Objectives 6.2 Introduction 6.3 All about the Supreme
6.4 Name &Form of the Supreme 6.5 Abode of the Supreme
6.6 Attributes of the Supreme 6.7 Action of benediction
6.8 Auspicious confluence (Now or never) 6.9 Summary
6.10 Glossary 6.11 Suggested reading

Chapter 7 : HEALING RELATIONSHIPS WITH THE SUPREME

7.1 Objectives 7.2 Introduction 7.3 Significance of the relationships
7.4 All relations with ONE 7.5 major relations 7.6 True and eternal relations
7.7 Benefits of various relationships 7.8 The timeless dimensions 7.9 Summary
7.10 Glossary 7.11 Suggested reading

Chapter 8 : RAJYOGA MEDITATION

8.1 Objectives 8.2 Introduction 8.3 Methods of Meditation
8.4 Rajyoga meditation with a difference 8.5 Five fold impact of Rajyoga meditation
8.6 Stages of Rajyoga meditation 8.7 Attainments of Meditation
8.8 Research studies on meditation 8.9 Summary
8.10 Glossary 8.11 Suggested reading

Chapter 9 : SPIRITUAL LIFE STYLE

9.1 Objectives 9.2 Introduction 9.3 Early morning meditation
9.4 Regular spiritual study 9.5 Authentic life style 9.6 Satwic Diet
9.7 Selfless service 9.8 Review of Personal Progress
9.9 Summary 9.10 Glossary 9.11 Suggested reading

Chapter 10 : EXERCISES FOR PRACTICE

Quiet reflection- Practice introversion-Being an observer-Stand back and observe -Self awareness (Soul consciousness)-Experiencing Body free stage-Reflect on original qualities-Visualize the Divine-Think attributes of the Supreme-Developing a living relationship-Surrender to God-Create Good wishes for all-Visualization in Meditation: Orbs of Light- The forest-The Balloon

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./ B.Com./B.Sc

Semester – V

Course	Subject	Subject Code
B.Sc(Microbiology)	**Basic Computer Information Technology-I**	**FC-501/2**

Unit – I

INTRODUCTION TO COMPUTER ORGANIZATION –I :- History of development of Computer system concepts. Characteristics, Capability and limitations. Generation of computer. Types of PC's Desktop. Laptop, Notebook. Workstation & their Characteristics.

कम्प्यूटर ऑर्गनाईजेशन का परिचय – I

कम्प्यूटर का इतिहास, कम्प्यूटर सिस्टम विचारधारा, विशेषताएं, योग्यता एवं सीमाएं, कम्प्यूटर की पीढ़ियां, पी.सी. के प्रकार, डेस्कटॉप के प्रकार, लेपटॉप के प्रकार, नोटबुक, वर्कस्टेशन आदि की विशेषताएं।

Unit – II

INTRODUCTION TO COMPUTER ORGANIZATION –II :-Basic components of a computer system Control Unit, ALU. Input/ Output function and Characteristics, memory RAM, ROM, EPROM, PROM.

कम्प्यूटर ऑर्गनाईजेशन का परिचय – II

कम्प्यूटर सिस्टम के आधार उपकरण, कंट्रोल युनिट, ए.एल.यू. इनपूट/ आउटपुट फंक्शन और विशेषताएं, मेमोरी रेम, रोम, ईपी रोम, पी रोम, और अन्य प्रकार की मैमोरी।

Unit – III

INPUT & OUTPUT DEVICES :- Input Devices : Keyboard, Mouse, Trackball. Joystick, Digitizing tablet, Scanners, Digital Camera, MICR, OCR, OMR, Bar-code Reader, Voice Recognition, Light pen, Touch Screen. **Output Devices:** Monitors Characteristics and types of monitor, Video Standard VGA, SVGA, XGA, LCD Screen etc. Printer, Daisy wheel, Dot Matrix, Inkjet, Laser, Line Printer. Plotter, Sound Card and Speakers.

इनपुट तथा आउटपुट डिवाईसेस

इनपुट डिवाइस : कीबोर्ड, माउस, ट्रेकबॉल, जॉयस्टिक, डिजिटाईजिंग टेबलेट, स्केनर्स, डिजिटल केमरा, एमआईसीआर, ओसीआर, ओएमआर, बार कोड रीडर, आवाज को पहचानने वाला, लाइट पेन, टच स्क्रीन।

आउटपुट डिवाइस : मॉनीटर की विशेषताएं एवं मोनीटर के प्रकार, वीडियों स्टैण्डर्ड VGA, SVGA, XGA, LCD स्क्रीन आदि, प्रिंटर्स, डेजी व्हील, डॉट मैट्रिक्स, इंकजेट, लेजर, लाईन प्रिंटर, प्लोटर, साउंड कार्डस एवं स्पीकर्स।

Unit – IV

STORAGE DEVICES :- Storage fundamental primary Vs Secondary. Various Storage Devices magnetic Tape. Cartridge Tape, Data Drives, Hard Drives, Floppy Disks, CD, VCD, CD-R, CD-RW, Zip Drive, DVD, DVD-RW.

स्टोरेज डिवाईजे़स

स्टोरेज फंडामेंटल्स प्राईमेरी विरुद्ध भिन्न स्टोरेज डिवाईजेस मेग्नेटिक टेप, काटरिज टेप, डाआ ड्राईव्स, हार्ड डिस्क ड्राईव्स, फ्लोपी डिस्कस, सी.डी. वी.सी.डी. सी.डी.–आर.सी.डी.– आर. डब्ल्यू, जीप ड्राईव, डी.वी.डी., डी.वी.डी., – आर. डब्ल्यू।

Unit – V

INTRODUCTION TO OPERATING SYSTEM :- Introduction to operating systems, its functioning and types. basic commands of dos & Windows operating System.

Disk Operating System (DOS)

- Introduction, History and Versions of DOS.

DOS Basics

- Physical Structure of disk, Drive name, FAT, file & directory structure and naming rules, booting process, DOS system files.

DOS Commands

- Internal DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE etc.
- External CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS etc.

ऑपरेटिंग सिस्टम का परिचय

ऑपरेटिंग सिस्टम का परिचय, उसके लक्षण एवं प्रकार, डॉस एवं विन्डोस का मूल कमांड। ऑपरेटिंग सिस्टम : डिस्क ऑपरेटिंग सिस्टम (DOS), परिचयए इतिहास एवं वर्ज़न्स ऑफ डॉस।

डॉस बेसिक्स –

फिजीकल स्ट्रक्चर ऑफ डिस्क, ड्राईव नेम, फेट, फाईल एवं डायरेक्ट्री स्ट्रक्चर एवं नेमिंग नियम, बूटिंग प्रक्रिया, डॉस सिस्टम फाईल्स।

डॉस कमाण्डस–

- आंतरिक कमाण्डस DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE आदि।
- बाह्य कमाण्डस CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS आदि।

Books Recommended-

1- डॉ. एस. के. विजय, डॉ. पकंज सिंह : कम्प्यटूर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल

2- डॉ. पकंज सिहं कम्प्यूटर अध्ययन, राम प्रसाद एडं संस

Course	Subject	Subject Code
B.Sc(Microbiology)	**Basic Computer Information Technology-I**	**FC 501/2**

PRACTICALS

DOS :

- DOS commands : Internal & External Commands.
- Special batch file : Autoexec, Bar Hard disk setup.

Windows 98:

- Desktop setting : New folder, rename bin operation, briefcase, function. Control panel utility.
- Display properties: Screen saver, background settings.

Ms-Word:

- Creating file: save, save as HTML, Save as Text, template, RTF Format.
- Page setup utility: Margin settings, paper size setting, paper source, layout.
- Editing: Cut, paste special, undo, redo, find, replace, goto etc.
- View file: page layout, Normal Outline, master document, ruler header, footer, footnote, full screen.
- Insert: break, page number, symbol, date & time, auto text, caption file, object, hyperlink, picture etc.
- Format: font, paragraph, bullets & numbering, border & shading, change case, columns.
- Table : Draw label, insert table, cell handling, table auto format, sort formula.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Microbiology)	BIOINFORMATICS , BIOSTATISTICS & INDUSTRIAL MICROBIOLOGY	BMB 151

Unit I

Introduction to Bioinformatics

Bioinformatics- Definition and relation to molecular biology, Potential of bioinformatics, Application of bioinformatics.

Databases

Nucleic acid and Protein databases, Structure databases, Enzyme databases, Specialized (organism and species) databases.

Unit II

Biostatistics

Measure of central tendency- Mean, mode and median, Measure of dispersion- Standard deviation and Standard error, Diagrammatic and graphic representation of frequency distribution.

Biostatistics II

Basic idea of probability- Addition and Multiplication laws, Test of significance- Chi square test, Normal distribution and departures from normality.

Unit III

Fundamentals of Industrial Microbiology

General concepts of industrial microbiology, Primary and secondary screening, Strain development strategies, Sterilization of fermentor, media and air.

Fermentor Design

Types of fermentations processes, Design of typical batch fermentor, Factors affecting fermentor design, Control of agitation, aeration, pH, temperature and dissolved oxygen, Types of fermentors.

Unit IV

Scale up and DSP

Inoculum development, Scale up of fermentation process, Raw material for media preparation, Harvesting and product recovery.

Unit V

Industrial production - I

Production of antibiotics- Penicillin and semi-synthetic penicillins, Production of enzymes- Amylase, Immobilization of enzymes and applications of immobilized enzymes.

Industrial production – II

Production of solvent- Ethanol, Production of Vitamins- Cyanocobalamin, Production of Organic Acids- Acetic Acid, Production of Amino Acids- Glutamic Acid,

Recommended Books (Semester-V)

1. Bioinformatics, Author- Baxevanis.
2. Bioinformatics, Author- Higgins and Taylor.
3. The Internet and the New Biology: Tools for Genomic and Molecular Research, Author- Peruski and Peruski.
4. Functional Genomics- A Practical Approach, Author- Mark Schena.
5. Principles of Biostatistics, Authors- Pagano *et al.*

6. Introduction to Biostatistics, Authors- Forthoter and Lec.
7. Text of Microbiology, Author- Ananthanarayanan and Panikar.
8. Medical Microbiology, Vol. 1 : Microbial Infection, Vol. 2 : Practical Medical Microbiology, Authors- Mackie and McCartney.
9. Epidemiology and Infections, Author- Smith
10. Lecture Notes in Immunology, Author- I.R. Todd
11. Microbiology in Clinical Practice, Author- D.C. Shanson.
12. Diagnostic Microbiology, Authors- Baron, Peterson and Finegold.

YEAR-III SEMESTER-V
LIST OF EXPERIMENTS

1. Examination of urine – Physical, chemical, microscopic and bacteriological.
2. Isolation and identification of Gram positive bacteria
(a) *Staphylococcus sp.*
(b) *Streptococcus sp.*
3. Isolation and identification of Gram positive bacteria
a. *E. coli*
b. *Proteus sp.*
c. *Salmonella sp.*
4. Antibiotic sensitivity test by disc diffusion technique.
5. Isolation of antibiotic resistant mutants by gradient plate technique.
6. Measure of central tendencies- Mean, Mode and Median.
7. Explore NCBI.
8. To read GenBank entries.
9. To read SWISSPROT entries.
10. To perform sequence similarity search using BLAST.
11. To perform multiple sequence alignment using Clustal W.
12. To visualize PDBIB 1AJE with the help of RASMOL.

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B.Sc.(Microbiology)

Syllabus
Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	BASED ON INORGANIC, ORGANIC & PHYSICAL CHEMISTRY	BCH-151

UNIT-I

Limitations of valence bond theory, an elementary idea of crystal-field theory, crystal field split ting in octahedral, tetrahedral and square planar complexes, **Magnetic Properties of Transition Metal Complex** Types of magnetic behavior, methods of determining magnetic susceptibility, spin-only formula. L-S coupling, **Electron Spectra of Transition Metal Complexes** Types of electronic transitions, selection rules for d-d transitions ,spectroscopic ground states, spectro chemical series. Orgel-energy level diagram for d1 and d9 states, discussion of the electronic spectrum of $[Ti(H_2O)_6]^{3+}$ complex ion.

UNIT-II

Black-body radiation, Plank's radiation law, photoelectric effect, heat capacity of solids, Compton effect, wave function and its significance of Postulates of quantum mechanics , quantum mechanical operator, commutation relations, Hamiltonial operator, Hermitian operator, average value of square of Hermitian as appositive quantity, Role of operators in quantum mechanics, To show quantum mechanically that position and momentum cannot be predicated simultaneously, Determination of wave function & energy of a particle in one dimensional box, Pictorial representation and its significance.

UNIT-III

Introduction: Electromagnetic radiation, regions of spectrum, basic features of spectroscopy, **Rotational Spectrum** Diatomic molecules. Energy levels of rigid rotator (semi-classical principles),selection rules. .**Vibrational spectrum** Infrared spectrum: Energy levels of simple harmonic oscillator, selection rules, pure vibrational spectrum, effects of an harmonic motion and isotopic effect on the spectra., idea of vibrational frequencies of different functional groups. **Raman Spectrum**: Concept of polarizibility, pure rotational and pure vibrational Raman spectra of diatomic molecules, select in rules, Quantum theory of Raman spectra.

UNIT-IV

Classification and nomenclature. Monosaccharides, mechanism of osazone formation, inter conversion of glucose and fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides. Erythro and threo diastereomers. Conversion of glucose in to mannose. Formation of glycos ides, ethers and esters. Determination of ring size of glucose and fructose. Open chain andcyclic structure of D(+)-glucose & D(-) fructose. Mechanism of mutarotation. Structures of ribose and deoxyribose. An introduction to disaccharides (maltose, sucrose and lactose) and polysaccharides (starch and cellulose) without involving structure determination.

UNIT-V

Organomagnesium compounds: the Grignard reagents-formation,structure and chemical reactions. Organozinc compounds: formation and chemical reactions. Organolithium compounds: formation and chemical reactions.

Books:

1. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshow, Pergamon.
3. Chemistry of the Elements. N.N. Greenwood and A. Earnshow, Pergamon.
4. P.W. Atkins , Physical Chemistry, ELBS.

(Practical)

SECTION – I (Inorganic)

Semimicro qualitative analysis of mixture containing not more than four radicals (including interfering, Combinations and excluding insoluables):

Pb^{2+}, Hg^{2+} , Hg_2^{2+}, Ag^+, Bi^{3+}, Cu^{2+}, Cd^{2+}, As^{3+} , Sb^{3+}, Sn^{2+}, Fe^{3+} ,Cr^{3+}, Al^{3+} , Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+}, Mg^{2+}, NH_4,CO_3^{2-}, S^{2-}, SO_3^{2-}, $S_2O_3^{2-}$, NO_2^-, CH_3COO^-, Cl^-, Br^-, I^-,NO_3^-, SO_4^{2-},$C_2O_4^{2-}$, PO_4^{3-}, BO_3

Section-B (Physical)

1. To determine the strength of the given acid solution (mono and dibasic acid) conductometrically.
2. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometrically
3. To determine the strength of given acid solution (mono and dibasic acid)/$KMnO_4$ – Mohr salt potentiometrically.

Section-C (Organic)

1. Laboratory Techniques

(a) Steam distillation (non evaluative)Naphthalene from its suspension in water Separation of *o*-and *p*-nitrophenols

(b) Column chromatography (non evaluative)

Separation of fluorescein and methylene blueSeparation of leaf pigments from spinach leaves

2. Chromatography Method

Determination of Rf values and identification o f organic compunds

(a) Separation of green leaf pigments (spinach leaves may be used)by paper chromatographic method

(b) Separation of a mixture of coloured organic compounds using common organic solventsby TLC.

3. Synthesis of the following organic compounds:

1. To prepare p-bromoaniline from p-bromoacetanilide.
2. To prepare m-nitroaniline from m-dinitrobenzene.

Fifth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Plant physiology and biochemistry	BSB 151

UNIT –1

Plant Water Relations: Properties of water, Importance of water in plant life, Diffusion, Osmosis & Osmotic relation to plant cell, Water Absorption, Ascent of Sap, Essential macro & micronutrients and their role. Transpiration: Structure & Physiology of Stomata, Mechanism of Transpiration, Factors affecting the rate of transpiration.

UNIT –2

Photosynthesis: Chloroplast, Photosynthetic pigments, Red drop, Emerson's effect, Concept of two Photosystems, Light reaction, Dark reaction - Calvin cycle, Hatch-Slack cycle, CAM cycle, Factors affecting rate of photosynthesis & Photorespiration.

UNIT –3

Respiration: Mitochondria, aerobic and anaerobic respiration, Respiratory coefficient, mechanism of respiration - Glycolysis, Kreb's cycle, Pentose phosphate pathway, Electron transport system, Factors affecting rate of respiration, Redox potential and theories of ATP synthesis.

UNIT – 4

Definition, classification and chemical structure: Monosaccharide, disaccharide, oligosaccharide and polysaccharides; Amino acids, essential and non essential amino acids; Lipids, saturated and non saturated fatty acids.

Classification, nomenclature and characteristics of Enzymes, Concept of holoenzyme, apoenzyme, co-enzyme and co-factors, mode & mechanism of enzyme action, Factors affecting enzyme activity. Plant Harmones, mode of action of Auxins, Gibberellins, Cytokinin and Abscissic acid.

UNIT – 5

Genetic Engineering: Tools and techniques of recombinant DNA technology; cloning vectors; genomic and cDNA library; transposable elements; gene mapping and chromosome walking. Biotechnology: Functional definition; basic aspects of plant tissue culture; cellular totipotency, differentiation and morphogenesis biology of Agrobacterium; vectors for gene delivery and marker genes; salient achievements in crop biotechnology.

Practical Exercises + Scheme (Marks- 50)

Question 1- 20

1- Preparation of solution of specific Normality, Molal and Molar solutions.

2- Exercises related to osmosis and osmotic relation.

3- Exercises related to Transpiration.

4- To separate Plastidial pigments by Paper Chromatography.

5- To perform the exercise of Photosynthesis & Respiration.

6- To perform biochemical test for Carbohydrate, Lipid and Protien.

7- To extract Enzyme for any plant part and demonstrate its activity.

(Any two experiments from above mentioned list)

Question 2: Comment on any technique related to Biotechnology- 05

Spotting- 10

Viva- voce- 5

Practical Record- 10

SUGGESTED READINGS:--

1• David, L. N. and Michael, M. C. 2000. Lehninger principle of Biochemistry, Macmillan worth Pub. New York, USA.

2• Gangulee, H.C.,Das, K.S., Datta, C. and Sen, S. 2007. College Botany Voll.I, New Central Book Agency (P) Ltd. Kolkata, 700009.

3• Hopkins, W.G. 1995. Introduction of Plant physiology Pub. John wiley and sons New York.

4• Jain, V.K. 1974. Fundamentals of Plant Physiology, S. Chand & Compnay.

5• Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.

6• Taiz & Zeiger, E. 1998. Plant Physiology. Sinauer associates, Inc. Pub. Massachusetts U.S.A.

7• Verma, S.K. & Verma, M.A. 1995. Text book of Plant Physiology & Biotechnology. S. Chand & Company.

8• Verma, V. 1995. Plant Physiology, Emkey Pub.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	MICROBIOLOGY AND BIOCHEMISTRY	BSZ 151

Unit-1

Classification of microorganisms- General characteristics of Bacteria, Virus, Yeast. Bacteria-Morphology, Bacterial cell structure, Motility, Nutrition and Reproduction. Virus-discovery-Morphology, Classification, phages and life cycle. Yeast-Morphology, cell structure, Multiplication, phages and cycle.

Unit-2

Morphology of water, air soil and sewage. Water-Microoorganisms of water, total bacterial count. Air-Microorganisms in soil, nitrogen cycle. Sewage-Composition of sewage, treatment of sewage by microorganisms.

Unit-3

Food borne diseases- Microbial food poisoning by Salmonella and Clostiridium botulinum (Botulism).Measures to prevent microbial food poisoning. Food infection-Food borne diseases- Diarrhea, Dysentery, Typoid and Cholera.Water borne diseases-Hepatitis, Gastro enteritis, Camphlo bacter-diarrhea, Gardia lamblia, Cryptosporidiosis cholera. Air borne diseases-Common cold, Tuberculosis, Pneumonia, Diphtheria.

Unit-4

Biochemistry- Definition and its importance, Physio- chemical forces acting on the living body –a) Definition of pH and its determination, Maintenance of pH of blood.
b)(Definition of osmosis, abnormality in edema and dehydration.
Nucleic acids, structure and classification.

Unit-5

Carbohydrates, lipids, Amino acids & proteins-Classification, structure and their function. Metabolism-Glycolysis-TCA cycle - Electorn Transport chain, Urea cycle Deamination, Oxidation of fatty acids.

REFERENCES:

Microbiology- Pelzer.
Biology of Microorganism- Madigan-Brock
Microbiology Lab manual – Capachim.
Microbiology fundamentals and application- Atlas.R.M.
Principles of Biochemistry A.L. Lehninger, D.L. Nelson& M.M. Cox (1993) Worth publishes New York.
Biochemistry by L.Stryer (1994) freeman & co., Newyork.
Biochemistry by Zubay (1998) Macmillan publishers & co., New York.

MAJOR PRACTICALS:

1.Estimation of dissolved oxygen content in the given water sample (Wrinklers method).

2. Estimation of salinity and pH in given water sample.

3. Plankton study –Identification and description of any five marine planktons.

MINOR PRACTICALS:

4. Examination of yeast, mould, protozoa and patho genic bacteria.

5. Estimation of urine sugar.

6. Blood grouping.

7. Problems on calculation of Mean, median, mode.

Spotters

Developmental Biology-Slides

Slides of mammalian sperm and Ovum

Slides of different developmental stages of chick embryos (24, 48, 72, 96 hrs)

Slides of blastula and gastrula of frog (morula, early gastrula, yolk plug stage, late gastrula)

Placenta of Sheep / Pig/ Rat.

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SCHEME

Semester – VI

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-601/1	Core	**Values & Spirituality**	10	4	40	14	-	-	50
2	FC-601/2	Core	**Basic Computer Information Technology-I**	10	4	40	14	-	-	50
3	BMB 161	Core	**Analytical Microbiology & Applied Microbiology**	20	8	80	27	50	17	150
4	BCH 161	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 161	Optional	Plant Ecology, Biodiversity And Phytogeography	20	8	80	27	50	17	150
	BSZ 161	Optional	Biostatistics And Computer Applications & Basic Animal Biotechnology	20	8	80	27	50	17	150
			Total	**80**	**32**	**320**	**109**	**150**	**51**	**550**

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B.C.A./B.A./B.Com./B.Sc
Semester – VI

Course	Subject	Subject Code
B.Sc(Microbiology)	भाषा कौशल एवं व्यक्तित्व विकास	FC-601/1

इकाई –1

1. भारतीय संस्कृति
2. भारतीय समाज व्यवस्था
3. सभ्यता एवं संस्कार
4. वैश्विक चेतना
5. समन्वयीकरण (भारतीय एवं अंतर्राष्ट्रीय संदर्भ में)

bdkbZ & 2

1. धर्म
2. न्याय
3. दर्शन
4. नीति
5. साहित्य

bdkbZ & 3

1. संचार संसाधन : सम्पर्क के नए क्षितिज
2. समाचार पत्र
3. भारतीय प्रेस परिषद्
4. रेडियों
5. दूरदर्शन

bdkbZ & 4

1. सिनेमा
2. रंगमंच
3. संगीत
4. चित्र, मूर्ति, स्थापत्य कला
5. शिल्प कला

bdkbZ & 5

1. कम्प्यूटर
2. दूरभाष : विज्ञान की सौगात
3. मंत्र (कहानी) : प्रेमचंद
4. मातृभूमि (कविता) : मैथिलीशरण गुप्त
6. साहित्यकार का दायित्व : डॉ. प्रेम भारती

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RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

B.C.A./B.A./B.Com./B.Sc

Semester – VI

Course	Subject	Subject Code
B.Sc(Microbiology)	Basic Computer Information Technology-II	FC-601/2

Unit-I

Word Processing: Word

- Introduction to word Processing.
- MS Word: features, Creating, Saving and Operating Multi document windows, Editing Text selecting, Inserting, deleting moving text.
- Previewing documents, Printing document to file page. Reduce the number of pages by one.
- Formatting Documents: paragraph formats, aligning Text and Paragraph, Borders and shading, Headers and Footers, Multiple Columns.

Unit-II

Introduction to Excel

Excel & Worksheet:

- Worksheet basic.
- Creating worksheet, entering data into worksheet, heading information, data text, dates, alphanumeric, values, saving & quitting worksheet.
- Opening and moving around in an existing worksheet.
- Toolbars and Menus, keyboard shortcuts.
- Working with single and multiple workbook coping, renaming, moving, adding and deleting. coping entries and moving between workbooks.
- Working with formulas & cell referencing.
- Autosum.
- Coping formulas
- Absolute & Relative addressing.

Unit-III

Introduction to Power Point

- Features and various versions.
- Creating presentation using Slide master and template in various colour scheme.
- Working with slides make new slide move, copy, delete, duplicate, lay outing of slide, zoom in or out of a slide.
- Editing and formatting text: Alignment, editing, inserting, deleting, selecting, formatting of text, find and replace text.

Unit-IV

Power Point– II

- Bullets , footer, paragraph formatting, spell checking.
- Printing presentation Print slides, notes, handouts and outlines.
- Inserting objects Drawing and Inserting objects using Clip Arts picture and charts.
- Slide sorter, slide transition effect and animation effects. Presenting the show making stand alone presentation, Pack and go wizards.

Unit – V

Evolution, Protocol, concept, Internet, Dial-up connectivity, leased line, VSAT, Broad band, URLs, Domain names, Portals. E-mail, Pop & web based Email. Basic of sending and receiving Emails, Email & Internet Ethics, Computer virus, Antivirus software wage, Web Browsers.

Books Recommended-

1. डॉ. एस. के. विजय, डॉ. पकंज सिंह : कम्प्यटूर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल
2. डॉ. पकंज सिहं कम्प्यूटर अध्ययन, राम प्रसाद एडं संस

Course	Subject	Subject Code
B.Sc(Microbiology)	Basic Computer Information Technology-I	FC 601/2

Practical

Ms-Power Point:

Creating new slide, formatting slide layout, slide show & sorter, Inserting new slide, slide no., date, time, chart, formatting slide, tool operation.

List of suggested practical work:

- Under standing of a dial up connection through modern.
- Configuring a computer for an e-mail and using outlook Express or Netscape Messenger.
- Registration an e-mail address.
- Understanding of e-mail drafting.
- Understanding of address book maintenance for e-mail.
- Understanding of different mail program tools.
- Send and receive functions of e-mail.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc (Microbiology)	**ANALYTICAL MICROBIOLOGY & APPLIED MICROBIOLOGY**	**BMB 161**

Unit I

Bioassays

Bioassay of growth supporting substances- Amino acids and Vitamins, Bioassay of growth inhibiting substances- Antibiotics, Automation of bioassay.

Quality Control

Quality control tests- Sterility testing, Microbial Limit Test (MLT), Pyrogen testing (LAL test), Minimum Inhibitory Concentration(MIC), FDA and Good Manufacturing Practices, Quantitative and qualitative analysis of food, milk, water and sewage.

Unit II

Colorimetry and Spectrophotometry

Lambert – Beer's Law, Ultraviolet, Visible, Infra red and Fluorescence spectroscopy, Atomic absorption, Raman spectrum, X-ray Crystallography and NMR.

Separation Techniques- I

Chromatography- Principle, Types of chromatography- Paper, Thin layer, Column, Ion exchange and Gas chromatography.

Unit III

Separation Techniques -II

Electrophoresis- Principle and working, Agarose gel, native PAGE and SDS-PAGE, Principle, working and applications of centrifuge.

Unit IV

Microorganisms in Agriculture

Bacteria and fungi as biopestcides**,** Genetically modified crops containing insecticidal genes, Biofertilizers- Nitrogen fixers, PSB and Mycorrhiza, Fuel from microorganisms- Biogas technology, Microbial hydrogen production, Concept of gasohol.

Unit V

Pharmaceutical Biotechnology

Genetically engineered microorganisms, Production of heterologous proteins- Insulin, Growth hormones, Interleukins and t plasminogen activator, Recombinant vaccines.

Food from Microbes

Dairy products- Cheese, Butter, Yogurt, Microorganisms as food- SCP, *Spirullina* and Mushroom, Indian and Oriental fermented foods.

Recommended Books (Semester-VI)

1. Textbook of Industrial Microbiology, Author- A. H. Patel.
2. Industrial Microbiology, Author- L. E. Cassida
3. Industrial Microbiology, Author- G. Reed.
4. Industrial Microbiology, Author- Agarwal And Parihar.
5. Biology of Industrial Microorganisms. A.L. Demain.
6. Principles of Fermentation Technology, Authors- Standbary, Whitaker and Hall.
7. Principles of Physical Biochemistry, Authors- Van Holde *et.al*.
8. Biochemistry of Nucleic Acids, Authors- Adams et. al.
9. Bioseparation: Principles and Techniques, Author- B. Sivasankar.
10. Protein Analysis and Purification, Authors- I.M. Rosenberg.

YEAR-III SEMESTER-VI

LIST OF EXPERIMENTS

1. Isolation of antibiotic producer from soil sample.
2. Isolation of amylase producer from soil sample.
3. Estimation of soil microflora.
4. Qualitative and quantitative examination of Food.
5. Qualitative and quantitative examination of Milk.
6. Qualitative and quantitative examination of Water.
7. Qualitative and quantitative examination of Sewage.
8. Bioassay of penicillin.
9. Bioassay of vitamin.
10. Sugar estimation by Cole's Method.
11. Estimation of MIC.
12. Sterility testing of pharmaceutical products- injectibles, eye and ear drops.
13. Microbial Limit Test- Tablets and syrups.
14. Determination of Phenol coefficient.
15. Separation of amino acids by TLC.
16. Separation of sugars by Paper chromatography.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	Based on inorganic, Organic & Physical Chemistry	BCH-161

UNIT-I

Arrhenius, Bronsted – Lowry, the Lux – Flood, Solvent system and Lewis concepts of acids & bases, relative strength of acids & bases, Concept of Hard and Soft Acids & Bases. Symbiosis, electronegativity and hardness and softness.

Bioinorganic Chemistry

Essential and trace elements in biological processes, metalloporphyrins with special reference to haemoglobin and myoglobin. Biological role of alkali and alkaline earth metal ions with special reference to Ca2+. Nitrogen fixation.

UNIT-II

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of photochemistry: Grotthus-Drapper law, Stark-Einstein law (law of photochemical equivalence) Jablonski diagram depiciting various processes occurring in the excited state, qualitative description offluorescence, phosphorescence, non-radiative processes (internal conversion,intersystem crossing), quantum yield, photosensitized reactions-energy transfer processes (simple examples).

UNIT-III

Statement and meaning of the terms – phase component and degree of freedom, thermodynamic derivation of Gibbs phase rule, phase equilibria of one component system – Example – water and Sulpher systems. Phase equilibria of two component systems solid-liquid equilibria, simple eutectic Example Pb-Ag system, desilerisation of lead.

UNIT-IV

Introduction: Molecular orbital picture and aromatic characteristics of pyrrole, furan, thiophene and pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine derivatives. **Heterocyclic Compounds-II** Introduction to condensed five and six- membered heterocyclic. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fisher indole synthesis, Skraup synthesis and Bischler-Napieralski synthesis.

UNIT-V

Nomenclature, structural features, Methods of formation and chemical reactions of thiols, thioethers, sulphonic acids,sulphonamides and sulphaguanidine. Synthetic detergents alkyland aryl sulphonates. **Amino Acids, Peptides& Proteins**

Classification, of amino acids. Acid-base behavior, isoelectricpoint and electrophoresis. Preparation of β -amino acids. Structure and nomenclature of peptides and proteins. Classification of proteins. Peptide structure determination, end group analysis, selective hydrolysis of peptides. Classical peptide synthesis, solid–phase peptide synthesis. Structures of peptides and proteins: Primary & Secondary structure.

Books:

1. Ira N. Levine, Quantum Chemistry, Prentice Hall.
2. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
3. K.J. Laidler, Chemical Kinetics. McGraw-Hill.
4. Inorganic Chemistry, J.E. Huhey, Harpes & Row.
5. Valence, C. A. Coulson, Oxford University Press.

(Practical)

Section-B (Physical)

1. To determine the molecular weight of a non-volatile solute by Rast method.
2. To standardize the given acid solution (mono and dibasic acid) pH metrically.

Section-C (Organic)

2. Chromatography Method

Determination of Rf values and identification o f organic compunds

(a) Separation of green leaf pigments (spinach leaves may be used)by paper chromatographic method

(b) Separation of a mixture of coloured organiccompounds using common organic solventsby TLC.

3. Synthesis of the following organic compounds:

1.To prepare o-chlorobenzoic acid from anthranilic acid.

2.To prepare S-Benzyl-iso-thiouronium chloride from thiourea.

Book for practical:

1.Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.

2.Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M.Miller, Prentice Hall.

3.Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Sixth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	**Plant ecology, biodiversity and phytogeography**	**BSB 161**

UNIT – 1

Ecosystems: Structure and types, Biotic and Abiotic components, Trophic levels, Food chains, Food webs, Ecological pyramids, Energy flow; Biogeochemical cycles: Concept, Gaseous and Sedimentary cycles, Carbon, Nitrogen, Phosphorus and Sulphur cycle.

UNIT – 2

Ecological adaptations: Morphological, Anatomical and Physiological responses, Water adaptation (Hydrophytes, Xerophytes and Mesophytes), Temperature adaptation (Thermoperiodism and Vernalization),Light adaptation (Heliophytes and Sciophytes), Plant Succession: Causes, trends and processes, types of succession - Lithosere, Hydrosere and Xerosere.

UNIT – 3

Population Ecology: Distribution patterns, Density, Natality, Mortality, Growth curves, Ecotypes and Ecads; Community Ecology: Characteristics, Classification, Life forms.

Biodiversity: Basic concept, definition, Importance, Biodiversity of India, Hotspots, In situ and ex situ conservation, Endangered and threatened species, Red data book.

UNIT – 4

Soil: Physico-chemical properties, Soil formation, Development of Soil Profile, Soil classification, Soil composition, Soil factors; Pollution: Definition, Types & Causes; Global warming, Climate change and Ozone holes.

UNIT – 5

Phytogeography: Phytogeographical regions of India, Vegetation -types of Madhya Pradesh, Biosphere reserves, Sancturies and National parks of Madhya Pradesh, Natural resources – definition and classification of natural resources, Conservation and management of natural resources, Land resources management,Water resources management, Wet land resource management.

Practical Exercises + Scheme
(Marks- 50)

1-To determine the minimum size of Quadrat by species area curve method. 05
- To conduct exercise on Frequency, Density and Abundance.
2- Study of soil with reference to soil texture, water holding capacity, pH and test for Carbonate and Nitrate. 05
3-Preparation of slides of Xerophytic, Hydrophytic and Mesophytic plants. 10
4-To comment upon Phytogeographic region (model/ charts) and National Parks (Photographs). 05
5-Spotting- 10
6-Viva- voce- 5
7-Practical Record- 10

SUGGESTED READINGS:--

1• Banerjee, S.1998. Bio diversity conservation- Agrobotamica, Bikaner.
2• Kumar, U.K 2006. Bio diversity principles and conservation, Agrobios, Jodhpur.
3• Odum, E.P. 5Th ed. 2004 .Fundamentals of Ecology. Natraj Publisher, Dehradun.
4• Puri, G.S. 1960. Indian Forest Ecology.
5• Sharma, P.D. 7th ed. 1998.Ecology and Environment, Rastogi Publication, Shivaji Road. Meerut, 250002. India.
6• Shukla, R. S. & Chandel, P.S. 2006. A Text book of Plant Ecology.

RKDF UNIVERSITY, BHOPAL

B.Sc.(Microbiology)

Syllabus
Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Microbiology)	BIOSTATISTICS AND COMPUTER APPLICATIONS & BASIC ANIMAL BIOTECHNOLOGY	BSZ 161

Unit-1

Introduction-definition, date types – primary and secondary – Classification of data, Collection of data – tabular and graphical representation – Bar diagram, Pi diagram, Column graph, Histogram, Ogive curves.

Unit-2

Measures of central tendency – Mean, Mode and Median, Variance, Standard deviation, Standard error and Coefficient or variance. Simple Correlation, Simple Regression, Chi square test, student's – t- test.

Unit-3

Fundamentals of Computer: Classification, Computer organization, Input devices, processing unit, output devices, external storage devices, software, WWW, CONCEPT OF E-Mail.
Computer and its application to biology-Definition and scope of Bioinformatics - application and introduction to Biological data.

Unit-4

History of animal cell culture, Laboratory requirements for animal cell culture, Sterilization techniques.
Media used for animal cell culture, Types of cell culture (Primary and Secondary), Introduction to established cell lines, Stem cells.

Unit-5

Cryopreservation of cell cultures, application of animal cell cultures in production of therapeutics proteins. Hybridoma technology.
Gene transfer methods in Animals – Microinjection, Embryonic Stem cell gene transfer, Retrovirus & Gene transfer. Transgenic Animals, Animal propagation – Artificial insemination,

REFERENCES:

Introduction of Biostatistics and Computer Science – Y.I Parkar & M.G Dhanyagude NiraliPrakashan publishers, Pune.
Biostatistics by K.S. Negi ATIBS publications & distributiors, New Delhi.
Bishop O.N. Statistics for Biology. Boston, Hollghtan, Mifflin.
Introduction to Biostatistics by Pranab kumar, S.Chand company Ltd. New Delhi.